
Detecting Errors in AI-Generated Annotations: When and Why Semantic Neighbors Help

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Abstract

Large language models (LLMs) and vision-language models (VLMs) have emerged as efficient annotators for tasks such as generation and classification. While these models offer significant cost and speed advantages over human annotation, a critical challenge remains: existing self-evaluation methods, such as LLM-as-judge, often lack reliable reference-based calibration for error detection. We address this limitation by introducing **SAGE** (Semantic-Anchored JudGmEnt), a method that leverages semantically similar samples retrieved via k -nearest-neighbor as references for annotation verification. We provide a theoretical framework that derives a closed-form expression for the error detection AUROC, which can be decomposed into three factors: intrinsic separability, reference-induced mean shift, and noise reduction through averaging. This decomposition reveals *when* semantic neighbors help (when references are both semantically matched and correct) and *why* (by providing reference-based calibration that raises scores for correct annotations and lowers scores for incorrect ones). Experiments on LLM generation, VLM captioning, and classification tasks validate our theoretical framework: SAGE improves error detection when semantic neighbors provide reliable reference-based calibration, and our decomposition offers insights into when direct scoring or alternative strategies may be preferred. Our code is available at <https://github.com/dina-1205/SAGE>.

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1. Introduction

The emergence of large language models (LLMs) and vision-language models (VLMs) has transformed data annotation pipelines across machine learning. These models can now generate answers to questions, produce image captions, and perform classification tasks at scale, achieving in hours what previously required weeks of human effort. This paradigm shift from human annotators to model-based annotators (Lee et al., 2013) offers compelling advantages in cost and throughput, making it increasingly popular for building large-scale datasets.

However, AI-generated annotations are not infallible. LLMs may hallucinate facts (Ji et al., 2023), VLMs may misinterpret visual content (Zhou et al., 2023), and both can exhibit systematic biases. A natural approach for quality control is to use self-evaluation methods such as LLM-as-judge, but prior work has shown that these paradigms suffer from inherent limitations: models tend to be overconfident, exhibit self-preference bias, and lack reference-based calibration (Kadavath et al., 2022; Wei et al., 2023).

Our key insight is that *semantically similar instances can serve as natural reference points for calibration* (Zhou et al., 2003; Di Salvo et al., 2024). If a judge observes that similar instances have correct annotations with consistent patterns, it gains a reference frame to assess the target annotation. A correct annotation should align with these references, while an incorrect one should exhibit detectable discrepancies. This raises two fundamental questions: *when* do such semantic neighbors actually help, and *why*?

We introduce **SAGE** (Semantic-Anchored JudGmEnt), a method that retrieves semantically similar samples via kNN and uses them as references when judging annotation quality. Beyond the method, we provide a theoretical framework that derives a closed-form expression for the error detection AUROC, answering two fundamental questions:

- **When** do semantic neighbors help? We show that references are beneficial only when they are both semantically matched and themselves correct. We call such references “usable.”
- **Why** do semantic neighbors help? Our AUROC decomposition reveals that usable references amplify the score

gap between correct and incorrect annotations.

Our analysis yields practical insights: kNN retrieval outperforms random sampling because semantic similarity increases the probability of retrieving usable references. However, we also identify conditions under which semantic neighbors may fail to help, such as when retrieval quality is poor or when errors are correlated across semantically similar instances.

We validate our framework through experiments on diverse tasks: LLM generation (hallucination detection, factual accuracy), VLM captioning, and text/image classification. Our results demonstrate that SAGE improves error detection when semantic neighbors provide reliable reference-based calibration, and our theoretical analysis offers insights into when simpler alternatives may suffice.

Our contributions are summarized as follows.

- We propose SAGE, a simple yet effective method that uses kNN-retrieved semantic neighbors as references for detecting errors in AI-generated annotations.
- We develop a theoretical framework that derives a closed-form AUROC expression, precisely characterizing *when* semantic neighbors help (usability conditions) and *why* (calibration mechanisms).
- We validate our framework on LLM generation, VLM captioning, and classification tasks, showing that our theoretical analysis not only explains when SAGE succeeds, but also predicts when direct scoring may be preferred.

2. Related Work

2.1. LLMs and VLMs as Annotators

LLMs and VLMs are efficient alternatives to human annotators. Gilardi et al. (2023) showed ChatGPT can match or exceed crowd workers on text annotation (Wang et al., 2021), and Törnberg (2023) found the same for GPT-4 on political messages. Tan et al. (2024) surveyed LLM-based annotation and its scalability/quality trade-offs. For vision, LLaVA (Liu et al., 2023a) and InstructBLIP (Dai et al., 2023) enable large-scale captioning and VQA, and AnnoLLM improves quality via better prompting (He et al., 2024). Yet VLMs frequently hallucinate plausible but incorrect content (Liu et al., 2024a), and annotation quality varies by task and domain; models may exhibit systematic biases unlike human errors (Liu et al., 2025) that complicate automated dataset construction (Liu et al., 2024b). These concerns motivate reliable error detection.

2.2. LLM-as-Judge and Self-Evaluation

A natural approach to quality control is using LLMs as judges. Zheng et al. (2023) introduced this paradigm with MT-Bench and Chatbot Arena, showing over 80% GPT-

4 agreement with humans. Self-consistency (Wang et al., 2022; Liu et al., 2023b), which aggregates multiple sampled outputs, can also improve evaluation reliability.

However, LLM-based evaluation suffers from systematic biases. Wataoka et al. (2024) and Panickssery et al. (2024) demonstrated that LLMs exhibit self-preference bias, favoring their own outputs. Other documented biases include position bias and verbosity bias (Zheng et al., 2023). Li et al. (2024) surveyed these limitations. From a statistical perspective, Calderon et al. (2025) proposed hypothesis testing for LLM-human equivalence, and Pang et al. (2025b) modeled LLM rating errors via score transition matrices, while Wei et al. (2024) estimated factuality without gold-standard answers by expertise-weighting reference LLMs. These methods address aggregate-level assessment or score correction, but do not provide *reference-based calibration* for instance-level error detection, which our work supplies via semantic neighbors.

2.3. Label Error Detection and Data Cleaning

Classical methods for noisy label detection were designed for human annotators: Dawid & Skene (1979) proposed EM-based estimation of annotator error rates, with extensions for reliability and task difficulty (Snow et al., 2008). These assume independent errors, unlike the systematic biases of AI annotators.

For dataset-level detection, Northcutt et al. (2021a) introduced Confident Learning to identify mislabeled examples via the joint distribution of noisy and true labels, revealing pervasive errors in benchmark test sets (Northcutt et al., 2021b). Other approaches use weak supervision (Ratner et al., 2017) and prediction disagreement across models (Seung et al., 1992; Han et al., 2018).

Recent work recognizes that label noise is often instance-dependent rather than uniform (Zhu et al., 2021a). Cheng et al. (2021) and Wei et al. (2022b) showed that corruption patterns depend on input features and local structure, motivating noise-robust objectives (Wei & Liu, 2021; Wei et al., 2022a) and transition-matrix estimation beyond images (Zhu et al., 2022b). Zhu et al. (2022a) proposed model-agnostic criteria for identifying corrupted labels, later applied to improve dataset credibility for language models (Zhu et al., 2024) and token-level data selection for fine-tuning (Pang et al., 2025a); representation quality is itself a strong error signal (Cheng et al., 2023). Most relevant to our work, Di Salvo et al. (2024) proposed WANN, which uses feature embeddings and nearest neighbor voting to detect noisy labels, showing that semantic similarity gives strong error signals. Penquitt et al. (2025) explored semi-automated detection-plus-verification pipelines.

Our work differs by focusing on AI-generated annotations,

which exhibit systematic rather than random errors, and by providing a theoretical framework characterizing *when* semantic neighbors help (usability conditions) and *why* (calibration mechanisms).

2.4. Connection to In-Context Learning

SAGE places retrieved examples in the judge’s prompt, superficially resembling in-context learning (ICL) (Brown et al., 2020), and in particular retrieval-augmented ICL where demonstrations are selected by semantic similarity (Liu et al., 2022). The two differ in mechanism: ICL uses demonstrations to teach the model a task, with performance driven primarily by the *number* of examples; SAGE compares a target against a reference to assess its quality, with performance driven by reference *correctness* $\Pr(C=1)$ rather than quantity. Our reference-quality ablation (Appendix D.6) shows AUROC tracks effective $\Pr(C=1)$ monotonically, from 66.74% at $\Pr(C=1)=0.249$ up to an oracle upper bound of 81.26% at $\Pr(C=1)=1.0$. The k -ablation (Appendix D.4) further shows AUROC *peaks* at $k=9$ and slightly decreases at $k=15\text{--}21$, a quantity-saturating pattern incompatible with the more-is-better behavior of ICL.

3. Method

3.1. Problem Setup

Task formulation. Consider a scenario where an AI model (LLM or VLM) has been used to generate annotations for a dataset. Let $x \in \mathcal{X}$ denote an input instance: a question Q in LLM settings, or an image I or image-question pair (I, Q) in VLM settings. A generator model G_θ produces an annotation $\hat{a} = G_\theta(x)$, such as an answer, caption, or class label. We define the *correctness indicator*:

$$Y = \begin{cases} 1 & \text{if } \hat{a} \text{ is correct w.r.t. } x, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Our goal is *error detection*: given a collection of $(x, \hat{a})$ pairs *without* access to ground-truth labels Y , we aim to identify which annotations are likely incorrect ($Y = 0$).

Target set and reference pool. We introduce two sets that play distinct roles in our framework:

- **Target set** $D = \{(x_i, \hat{a}_i)\}_{i=1}^n$: The set of input-annotation pairs to be verified. Each pair in D receives a quality score indicating the likelihood that $\hat{a}_i$ is correct.
- **Reference pool** $P = \{(x'_j, \hat{a}'_j)\}$: The set of input-annotation pairs available as references during judgment. When evaluating a target pair, we can retrieve similar pairs from P to provide calibration context for the judge.

In general, P could be a separate high-quality dataset or coincide with D . Throughout this work, we focus on the

self-referencing setting where $P = D$: the target set serves as its own reference pool. When retrieving neighbors for a target $(x_i, \hat{a}_i)$, we exclude the target itself to avoid trivial self-matching. This setting is practically appealing as it requires no external labeled data.

Judging framework. To detect errors, we employ a judge model J_θ that evaluates each annotation and outputs a quality score $S \in \{0, 1, \dots, L\}$, where L denotes the maximum score level (e.g., $L = 5$) and larger scores indicate higher perceived quality. Following the LLM-as-judge paradigm, we use the *same model* for both generation and judgment: the generator G_θ and judge J_θ share the same parameters θ . This self-judgment setting is practically appealing as it requires no additional models, though our framework can be generalized to settings where distinct models are used to reduce systematic biases.

We formulate error detection as a ranking problem: given a scoring function that produces a real-valued score T for each annotation, we measure performance using the Area Under the ROC Curve (AUROC):

$$\text{AUROC} \triangleq \Pr(T_+ > T_-) + \frac{1}{2} \Pr(T_+ = T_-), \quad (2)$$

where $T_+ \sim T \mid Y = 1$ and $T_- \sim T \mid Y = 0$ are independent draws from correct and incorrect annotations, respectively. AUROC equals the probability that a randomly chosen correct annotation receives a higher score than a randomly chosen incorrect one. An AUROC of 0.5 corresponds to random guessing; higher values indicate better error detection.

3.2. Three Judging Strategies

A central question in designing an error detection system is: *what information should the judge have access to?* We consider three strategies that differ in whether and how they incorporate reference samples. Figure 1 provides an overview: the left panel shows concrete examples of in-context judgment for LLM and VLM tasks, while the right panel illustrates the three strategies. The progression from Strategy 1 to Strategy 3 reflects our core hypothesis: providing semantically relevant references enables better calibration.

Strategy 1: Direct scoring. The simplest approach has the judge evaluate the annotation $(x, \hat{a})$ in isolation:

$$S^{(1)} \triangleq J_\theta(x, \hat{a}), \quad T^{(1)} \triangleq S^{(1)}. \quad (3)$$

This is the standard LLM-as-judge paradigm. While straightforward, this approach lacks external calibration: the judge relies solely on its parametric knowledge, which may be incomplete or miscalibrated for the specific domain.

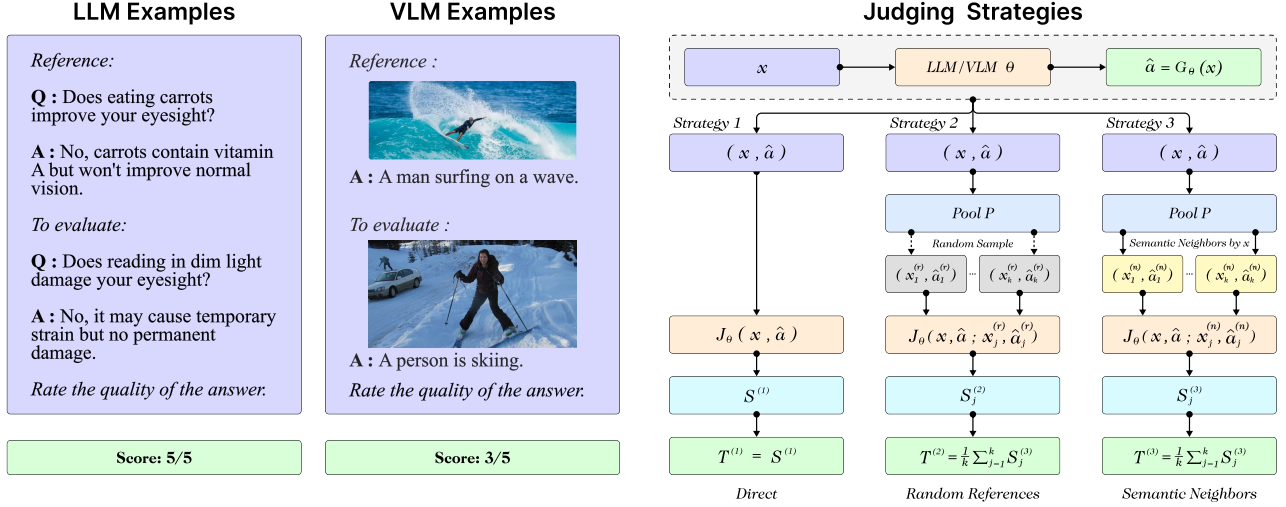


Figure 1. Overview of SAGE. **Left:** Examples of in-context judgment for LLM and VLM. The judge receives a reference pair $(x', \hat{a}')$ as context and evaluates the target annotation, denoted $J_\theta(x, \hat{a}; x', \hat{a}')$. **Right:** Three judging strategies. Given input x and generated answer $\hat{a}$: Strategy 1 judges directly without reference via $J_\theta(x, \hat{a})$; Strategies 2 and 3 provide references and average over k trials, differing only in how references are selected (random vs. kNN retrieval). All strategies use the same LLM/VLM as both generator and judge.

Strategy 2: Random reference scoring. To introduce reference-based calibration, we provide reference annotations in the judgment prompt. For each of k independent trials, we randomly sample a reference $(x_j^{(r)}, \hat{a}_j^{(r)})$ from the pool P and include it as context:

$$S_j^{(2)} \triangleq J_\theta(x, \hat{a}; x_j^{(r)}, \hat{a}_j^{(r)}), \quad T^{(2)} \triangleq \frac{1}{k} \sum_{j=1}^k S_j^{(2)}. \quad (4)$$

The final score averages over k trials, reducing variance. However, random references are unlikely to be semantically related to the target, limiting their utility as calibration anchors.

Strategy 3: Semantic-anchored judgment (SAGE). Our key insight is that *semantically similar* references provide more effective calibration. For each trial j , we retrieve the j -th nearest neighbor $(x_j^{(n)}, \hat{a}_j^{(n)})$ from P based on embedding similarity of the input x :

$$S_j^{(3)} \triangleq J_\theta(x, \hat{a}; x_j^{(n)}, \hat{a}_j^{(n)}), \quad T^{(3)} \triangleq \frac{1}{k} \sum_{j=1}^k S_j^{(3)}. \quad (5)$$

As illustrated in Figure 1 (left), when the neighbor addresses a semantically related input (e.g., a similar question or visually similar image) with a correct answer, it provides a meaningful reference point for comparison.

The SAGE algorithm. Algorithm 1 formalizes the complete procedure. The algorithm operates in two phases: (1) an *indexing phase* that computes embeddings for all samples in P using an encoder E and builds a kNN index for

Algorithm 1 SAGE: Semantic-Anchored Judgment

Require: Target set $D = \{(x_i, \hat{a}_i)\}_{i=1}^n$, reference pool P , encoder E , judge J_θ , number of neighbors k

Ensure: Quality scores $\{T_i\}_{i=1}^n$

// *Indexing phase: build kNN index*

Compute embeddings $\{E(x')\}$ for all $(x', \hat{a}') \in P$

Build kNN index $\mathcal{I}$ from embeddings

// *Scoring phase: evaluate each target*

for $i = 1$ **to** n **do**

$e_i \leftarrow E(x_i)$ // *embed target input*

$(x_1^{(n)}, \hat{a}_1^{(n)}), \dots, (x_k^{(n)}, \hat{a}_k^{(n)}) \leftarrow \text{kNN}(\mathcal{I}, e_i, k)$

// *retrieve*

for $j = 1$ **to** k **do**

$S_{i,j} \leftarrow J_\theta(x_i, \hat{a}_i; x_j^{(n)}, \hat{a}_j^{(n)})$

// *judge with neighbors*

end for

$T_i \leftarrow \frac{1}{k} \sum_{j=1}^k S_{i,j}$ // *average scores*

end for

return $\{T_i\}_{i=1}^n$

efficient retrieval; and (2) a *scoring phase* that, for each target, retrieves top- k neighbors and invokes the judge k times with each neighbor as context. The final score averages over the k judgments, reducing variance and aggregating calibration anchors from multiple references. The encoder E can be either the generator’s own embedding function or an external encoder; we use the generator’s embeddings in our main experiments for simplicity (see Appendix C for analysis showing that a unified external encoder achieves comparable or better performance).

3.3. Why Do Semantic Neighbors Help?

We now provide intuition for why semantic neighbors improve error detection.

The limitation of isolated judgment. Strategy 1 evaluates each annotation without external context. The judge relies entirely on its parametric knowledge, which may be incomplete or miscalibrated. Without a reference point, the judge cannot distinguish between confidently correct and confidently wrong predictions.

References as calibration anchors. Providing references enables *comparative judgment*: instead of assessing an annotation in absolute terms, the judge can evaluate it relative to similar cases. For a reference to be useful, two conditions must hold: (i) the reference should be *semantically relevant* to the target, enabling meaningful comparison; (ii) the reference annotation should be *correct*, providing a reliable anchor.

Random vs. semantic retrieval. Random references (Strategy 2) rarely satisfy condition (i): a randomly sampled reference is unlikely to address a semantically related input. The lack of relevance renders the comparison uninformative.

In contrast, kNN retrieval (Strategy 3) selects references based on embedding similarity, making condition (i) more likely to be satisfied. Moreover, when the reference pool has reasonable accuracy, semantically similar samples are likely to have correct annotations, particularly for correct targets, whose neighbors tend to share similar (correct) answers. This increases the probability that condition (ii) is also met.

The mechanism of improvement. When both conditions hold, the reference provides a calibration anchor: correct targets exhibit consistency with their neighbors, while incorrect targets reveal discrepancies. This differential effect widens the score gap between correct and incorrect annotations, directly improving AUROC. Section 4 formalizes this intuition and derives precise conditions under which semantic neighbors improve error detection.

4. Theoretical Analysis

We now formalize the intuitions from Section 3.3, deriving conditions under which semantic neighbors improve error detection.

4.1. Reference Usability

Recall from Section 3.3 that a reference is useful for calibration when two conditions hold: (i) semantic relevance and (ii) correctness. We now formalize these conditions as follows.

Definition 4.1 (Usable Reference). For a reference $(x_{\text{ref}}, \hat{a}_{\text{ref}})$ used to judge target $(x, \hat{a})$, define the *usability indicator* as

$$U \triangleq \mathbf{1}[M = 1 \wedge C = 1],$$

where the semantic match indicator $M = 1$ iff x_{ref} is semantically similar to x , and the correctness indicator $C = 1$ iff $\hat{a}_{\text{ref}}$ is correct for x_{ref} .

The *usability rate* for class $y \in \{0, 1\}$ under method $m \in \{2, 3\}$ (random and kNN references, respectively) is:

$$u_y^{(m)} \triangleq \Pr(U = 1 \mid Y = y). \quad (6)$$

This rate depends on both the retrieval strategy and the target’s correctness. A key insight is that kNN retrieval (Strategy 3) achieves higher usability rates than random sampling, because semantic neighbors are more likely to satisfy the match condition $M = 1$.

4.2. Modeling Assumptions

We first establish the score decomposition (Lemma 4.2), then state two distributional assumptions to enable tractable analysis. Full derivations are provided in Appendix B.

Lemma 4.2 (Additive score decomposition). *For method $m \in \{2, 3\}$ and class $Y = y$, the per-trial score admits the exact decomposition:*

$$S_{y,j}^{(m)} = \mu_y^{(m)} + \delta_y^{(m)} U_{y,j}^{(m)} + \epsilon_{y,j}^{(m)}, \quad (7)$$

where $U_{y,j}^{(m)}$ denotes the usability indicator (Definition 4.1) for the j -th reference, $\mu_y^{(m)} \triangleq \mathbb{E}[S \mid Y = y, U = 0]$ is the baseline mean, $\delta_y^{(m)} \triangleq \mathbb{E}[S \mid Y = y, U = 1] - \mathbb{E}[S \mid Y = y, U = 0]$ is the mean shift induced by a usable reference, and $\epsilon_{y,j}^{(m)}$ is zero-mean residual noise.

Proof Sketch. Since U is binary, the conditional mean $\mathbb{E}[S \mid Y = y, U]$ is necessarily affine in U . Defining δ_y as the conditional mean difference and ϵ as the residual yields the decomposition by construction. See Appendix B.3 for the full derivation. $\square$

The key modeling insight is that usable references shift scores *differentially* by class:

- For correct targets ($y = 1$): usable references provide consistent signals, *raising* scores. We expect $\delta_1^{(m)} > 0$.
- For incorrect targets ($y = 0$): usable references expose discrepancies, *lowering* scores. We expect $\delta_0^{(m)} < 0$.

This differential effect is the mechanism through which references improve discrimination.

Assumption 4.3 (Gaussian approximation). The averaged score $T^{(m)} = \frac{1}{k} \sum_{j=1}^k S_{y,j}^{(m)}$, conditional on $(Y = y, N_y^{(m)} = n)$, is approximately Gaussian:

$$T^{(m)} \mid (Y=y, N_y^{(m)}=n) \approx \mathcal{N}\left(\mu_y^{(m)} + \delta_y^{(m)} n/k, \sigma_{y,\text{avg}}^{2,(m)}\right), \quad (8)$$

where $N_y^{(m)} = \sum_{j=1}^k U_{y,j}^{(m)}$ counts usable references among k trials, and $\sigma_{y,\text{avg}}^{2,(m)}$ is the variance of the averaged residual noise (see Appendix B.5.2 for derivation).

Justification. By the Central Limit Theorem, the average of k bounded scores converges to Gaussian. In practice, $k \geq 5$ typically suffices. See Appendix B.4 for detailed conditions and failure modes.

Assumption 4.4 (Independent usability). The usability indicators $\{U_{y,j}^{(m)}\}_{j=1}^k$ are approximately independent with $\Pr(U_{y,j}^{(m)} = 1 \mid Y = y) = u_y^{(m)}$, so that $N_y^{(m)} \sim \text{Binomial}(k, u_y^{(m)})$.

Justification. For random references, independence holds by construction. For kNN references, independence is approximate under the standard noisy-label assumption that annotation errors are conditionally independent given the ground-truth label (Zhu et al., 2021b). See Appendix B.4 for detailed conditions and the Beta-Binomial relaxation.

4.3. Main Result: Closed-Form AUROC

Under these assumptions, we derive an exact closed-form expression for AUROC. Let $B_k(n; u) \triangleq \binom{k}{n} u^n (1-u)^{k-n}$ denote the binomial probability mass function.

Theorem 4.5 (Closed-form AUROC). *Under Lemma 4.2 and Assumptions 4.3–4.4, for $m \in \{2, 3\}$:*

$$\text{AUROC}^{(m)} = \sum_{n_1=0, n_0=0}^k B_k(n_1; u_1^{(m)}) B_k(n_0; u_0^{(m)}) \cdot \Phi\left(\frac{\Delta\mu^{(m)} + \frac{\delta_1^{(m)} n_1 - \delta_0^{(m)} n_0}{k}}{\sigma_{\text{tot}}^{(m)}}\right), \quad (9)$$

where $\Delta\mu^{(m)} = \mu_1^{(m)} - \mu_0^{(m)}$, $\sigma_{\text{tot}}^{(m)} = \sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}$, and $\Phi(\cdot)$ is the standard normal cumulative distribution function.

Proof sketch. We decompose AUROC as a mixture over usability counts via the law of total probability. Conditioning on $(N_1 = n_1, N_0 = n_0)$, the Gaussian assumption (Assumption 4.3) implies that the score difference $T_1 - T_0$ is normally distributed, reducing the pairwise comparison $\Pr(T_1 > T_0)$ to evaluating the standard normal CDF at the

normalized mean difference. The independence assumption (Assumption 4.4) provides Binomial mixture weights, and summing over all (n_1, n_0) pairs yields the closed form. See Appendix B.5 for details. $\square$

Single-probit approximation. When the variance of usability counts is small relative to score noise, the mixture concentrates around its mean, yielding a simpler expression:

Corollary 4.6 (Single-probit approximation). *Under mild concentration conditions (see Appendix B.5.5 for derivation),*

$$\text{AUROC}^{(m)} \approx \Phi\left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)} u_1^{(m)} - \delta_0^{(m)} u_0^{(m)}}{\sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}}\right). \quad (10)$$

This approximation has an intuitive signal-to-noise ratio (SNR) interpretation. The numerator decomposes AUROC improvement into three additive contributions:

1. $\Delta\mu^{(m)}$: **Intrinsic separability**: the judge’s baseline ability to distinguish correct from incorrect annotations, independent of references.
2. $\delta_1^{(m)} u_1^{(m)}$: **Score shift for correct targets**: positive when $\delta_1 > 0$ (usable references raise scores for correct annotations).
3. $-\delta_0^{(m)} u_0^{(m)}$: **Score shift for incorrect targets**: positive when $\delta_0 < 0$ (usable references lower scores for incorrect annotations).

4.4. Insights and Implications

The closed-form expressions yield several insights that formalize and extend the intuitions from Section 3.3. A more detailed discussion is provided in Appendix B.6.

Random vs. Direct: the role of noise reduction. Even when references are semantically unrelated, Random (Strategy 2) benefits from noise reduction through k -trial averaging: $\sigma_{\text{avg}}^{(2)} < \sigma^{(1)}$. Since $u_1^{(2)} \approx u_0^{(2)}$ are both small, the usability terms roughly cancel, and Random’s advantage comes primarily from the reduced denominator in (10). Thus, *Random should generally outperform Direct* when averaging provides meaningful variance reduction.

Why SAGE outperforms random references. From (10), AUROC improves when usability rates u_1, u_0 increase (assuming $\delta_1 > 0, \delta_0 < 0$). SAGE’s advantage over Random comes from higher usability:

- **Strategy 2 (Random)**: $\Pr(M = 1)$ is small, yielding low usability $u_1^{(2)} \approx u_0^{(2)} \approx \text{small}$. Improvement over Direct comes mainly from noise reduction.

- **Strategy 3 (SAGE):** kNN retrieval increases $\Pr(M = 1)$ by selecting references with high embedding similarity, so $u_1^{(3)}$ is expected to be greater than $u_1^{(2)}$. SAGE gains from *both* noise reduction and increased mean separation.

Asymmetry in usability rates. An interesting prediction is that $u_1^{(3)} > u_0^{(3)}$ in many settings. Intuitively, correct targets tend to have correct neighbors (similar inputs often yield similar correct answers), while incorrect targets may have a mix of correct and incorrect neighbors. This asymmetry means SAGE benefits correct annotations more than it penalizes incorrect ones, but both effects contribute positively to AUROC.

When does SAGE fail? The analysis reveals failure modes:

- **Poor retrieval quality:** If embedding similarity does not predict semantic relevance ($\Pr(M = 1 \mid \text{kNN})$ is low), SAGE degenerates toward random references.
- **Correlated errors:** If the same model generates all annotations, errors may cluster in semantic neighborhoods. This can reduce $u_0^{(3)}$ (neighbors of incorrect answers are also incorrect), limiting SAGE’s ability to detect errors via discrepancy.
- **Low pool accuracy:** If the reference pool has many errors, even semantically matched references may be incorrect, reducing both $u_1^{(m)}$ and $u_0^{(m)}$.

Empirical grounding of latent quantities. The quantities δ_y , u_y , and the residual correlation ρ in Eq. (10) are latent but estimable from observed scores. On TruthfulQA (Qwen3-8B, $k=9$), stratifying neighbor-trial scores by target correctness Y and reference correctness C yields $\hat{\delta}_1 = +0.47$ and $\hat{\delta}_0 = -0.16$, matching the predicted signs. The average pairwise Pearson correlation of within-sample score residuals gives $\hat{\rho} \approx 0.45$, implying an effective sample size $k_{\text{eff}} = k/(1 + (k-1)\rho) \approx 2$ at $k=9$. A controlled ablation that manipulates effective $\Pr(C=1)$ by selectively dropping correct neighbors confirms the monotonic AUROC dependence predicted by Eq. (10). Full estimation procedures and the keep-rate table are reported in Appendix D.6.

5. Experiments

We evaluate SAGE on a diverse set of annotation tasks spanning text classification, question answering, hallucination detection, and image captioning. Our experiments aim to answer: (1) Does SAGE improve error detection over reference-free baselines? (2) When do semantic neighbors help, and when do they fail? (3) Do the empirical patterns align with our theoretical predictions?

5.1. Experimental Setup

Tasks and datasets. We organize our evaluation into three categories:

- **LLM Generation:** We evaluate on HaluEval (Li et al., 2023) for hallucination detection and TruthfulQA (Lin et al., 2022) for factual accuracy assessment. These datasets specifically target common LLM failure modes.
- **VLM Captioning:** We use COCO-Caption (Chen et al., 2015) and Flickr30k (Young et al., 2014) for image captioning tasks, testing SAGE’s applicability to vision-language models.
- **Classification:** We use AG News (Zhang et al., 2015) for news topic classification, MMLU (Hendrycks et al., 2021) for multi-domain knowledge assessment, and CIFAR-10/ImageNet-1k for image classification.

Models. We evaluate three 8B-scale instruction-tuned models for each modality: LLaMA-3.1 (Grattafiori et al., 2024), Ministral (Mistral AI, 2024), and Qwen3 (Yang et al., 2025) for LLM tasks; InternVL3.5 (Wang et al., 2025), Qwen3-VL (Bai et al., 2025), and SAIL-VL2 (Yin et al., 2025) for VLM tasks. Each model serves as both generator and judge.

Baselines and variants. We compare three main scoring strategies:

- **Direct:** Strategy 1 with direct scoring without any reference (standard LLM-as-judge).
- **Random:** Strategy 2 with randomly sampled references from the reference pool.
- **SAGE:** Our semantic neighbor retrieval approach. For LLM tasks, we retrieve neighbors based on question embeddings; for VLM tasks, we use image embeddings. All embeddings are extracted from the generator model itself. We retrieve $k = 9$ neighbors and average scores over all trials.

Appendix C presents additional experiments with alternative retrieval strategies, including unified external encoders.

Evaluation metric. We report AUROC as defined in Eq. (2), measuring the probability that a correct annotation receives a higher quality score than an incorrect one. Higher AUROC indicates better error detection capability.

Ground truth for generation tasks. For generation tasks (HaluEval, TruthfulQA, COCO-Caption, Flickr30k), ground truth correctness labels are obtained by comparing generated answers against reference answers using Qwen3-Next-80B-A3B-Instruct as an external judge. This ensures that the ground truth assessment is independent of the 8B-scale models being evaluated.

Table 1. LLM Generation: Error Detection AUROC (%)

Dataset	Model	Direct	Random	SAGE
HaluEval	LLaMA-3.1	66.65	65.84	70.47
HaluEval	Ministral	57.61	65.34	67.26
HaluEval	Qwen3	59.59	61.57	65.05
TruthfulQA	LLaMA-3.1	58.57	59.00	58.15
TruthfulQA	Ministral	62.75	60.95	63.54
TruthfulQA	Qwen3	65.34	69.02	72.61

Table 2. VLM Captioning: Error Detection AUROC (%)

Dataset	Model	Direct	Random	SAGE
COCO	InternVL3.5	58.38	53.86	60.01
COCO	Qwen3-VL	55.80	56.85	55.66
COCO	SAIL-VL2	59.54	58.88	58.99
Flickr30k	InternVL3.5	62.04	58.19	64.16
Flickr30k	Qwen3-VL	61.17	62.09	62.60
Flickr30k	SAIL-VL2	62.93	60.94	64.50

Table 3. Text Classification: Error Detection AUROC (%)

Dataset	Model	Direct	Random	SAGE
AG News	LLaMA-3.1	59.76	54.33	62.44
AG News	Ministral	61.29	54.20	64.79
AG News	Qwen3	69.30	63.11	71.94
MMLU	LLaMA-3.1	56.29	56.81	57.48
MMLU	Ministral	51.14	53.46	52.84
MMLU	Qwen3	67.85	70.84	68.99

Table 4. Image Classification: Error Detection AUROC (%)

Dataset	Model	Direct	Random	SAGE
CIFAR-10	InternVL3.5	72.03	78.13	92.99
CIFAR-10	Qwen3-VL	84.55	83.02	94.35
CIFAR-10	SAIL-VL2	74.16	59.45	66.93
ImageNet-1k	InternVL3.5	61.55	68.66	75.40
ImageNet-1k	Qwen3-VL	67.00	66.86	65.71
ImageNet-1k	SAIL-VL2	58.47	62.55	68.52

5.2. Main Results

LLM generation results. Table 1 shows that SAGE achieves consistent gains on HaluEval across all models (+3.8 to +9.7 points over Direct), with the improvement especially pronounced for Ministral where Direct scoring performs poorly (57.61%). On TruthfulQA, SAGE helps for Ministral and Qwen but not LLaMA. We attribute LLaMA’s underperformance to two factors: (1) LLaMA’s weaker self-judgment capability compared to other models, as evidenced by its lower Direct scores, and (2) suboptimal neighbor retrieval using question-only embeddings. Table 6 in the appendix shows that switching to full-input retrieval (Self) improves LLaMA’s SAGE from 58.15% to 60.28%, outperforming both Direct (58.57%) and Random (59.00%). Overall, SAGE outperforms Direct in 5 out of 6 LLM configurations.

VLM captioning results. On Flickr30k (Table 2), SAGE consistently wins for all three VLMs (+1.4 to +2.1 points over Direct), showing that visual semantic neighbors provide useful calibration for caption quality assessment. On COCO, results are mixed: SAGE helps only for InternVL. For Qwen, all three methods achieve similar performance (55.66–56.85%); we note that Qwen’s caption accuracy on COCO is 85%, approximately 10 points higher than the other two models, suggesting error detection may have reached a ceiling effect where both δ_0 and δ_1 approach zero. For SAIL, the underperformance stems from suboptimal self-embedding retrieval; Table 7 shows that switching to Unified kNN improves SAIL’s SAGE from 58.99% to 61.45%, surpassing Direct (59.54%).

Text classification results. Table 3 shows that on AG News, SAGE consistently outperforms baselines across all

three models (+2.6 to +3.5 points over Direct). The 4-class topic structure yields high semantic match rates for kNN retrieval, aligning with our theory. Random scoring performs near chance level ($\sim 54\%$), confirming that semantic relevance drives improvement. On MMLU, SAGE only helps for LLaMA, while Random scoring wins for Ministral and Qwen. MMLU is inherently difficult for error detection: the diverse multi-domain structure reduces semantic match effectiveness, and we hypothesize that the δ terms may not have opposite signs across correct and incorrect annotations.

Image classification results. Table 4 shows SAGE achieves substantial gains on CIFAR-10 for InternVL and Qwen (AUROC $> 92\%$), consistent with our theoretical prediction: visually similar images in CIFAR-10’s 10-class structure yield high semantic match rates, increasing reference usability. SAIL-VL is an exception where Direct scoring wins. On ImageNet-1k, SAGE improves performance for 2 out of 3 models, though the larger label space (1000 classes) reduces the match rate advantage. Overall, SAGE achieves the best performance in 4 out of 6 configurations.

Connecting results to theory. Our theoretical framework (Section 4) predicts that SAGE improves AUROC when references are both semantically matched and correct (i.e., usable). The empirical patterns support this:

- **Strong gains on HaluEval, Flickr30k, and AG News:** These datasets have clear semantic clusters, yielding high match rates $\Pr(M = 1)$ and usability $u_1^{(3)}$.
- **When SAGE is not optimal:** In 7 out of 24 configurations, SAGE does not achieve the best AUROC. These cases have diverse causes:
 - *Suboptimal retrieval* (TruthfulQA LLaMA, COCO

SAIL): The default retrieval strategy finds poor neighbors; switching strategies in Tables 6 and 7 recovers SAGE’s advantage.

- *Ceiling effect* (COCO Qwen): High annotation accuracy (85%) leaves little room for error detection improvement, with both δ_0 and δ_1 approaching zero.
- *Task difficulty* (MMLU Ministral/Qwen): The diverse multi-domain structure makes error detection inherently difficult, and δ terms may not have opposite signs.
- *Embedding quality* (Qwen on ImageNet): Self-embeddings yield suboptimal neighbors; Table 7 shows Unified kNN improves SAGE from 65.71% to 75.29%, far surpassing Direct (67.00%).
- *Correlated errors* (SAIL-VL on CIFAR-10): Even with Unified kNN (67.35%), SAGE remains below Direct (74.16%), suggesting systematic errors where semantically similar images share incorrect labels.

5.3. Robustness Analyses

Stronger baselines. To isolate the contributions of the SAGE pipeline, we compare against three additional baselines on TruthfulQA $\times$ Qwen3-8B (full table in Appendix D.1, Table 8). **DS2** (Pang et al., 2025b) uses the same kNN retrieval as SAGE but bypasses the judge, isolating the judge’s contribution: SAGE (72.61%) outperforms DS2 (67.10%) by +5.51 pp. The most informative comparison is **Self-consistency (judge)** (68.58%) versus **Random** (69.02%): both make k judge calls, but only Random provides external reference context. Random wins narrowly, confirming that external context (not repeated scoring) drives the improvement. Two further generation-stage confidence baselines (Logprob, Self-consistency over embeddings) fall near or below chance on TruthfulQA.

Cross-model evaluation. Using the same model as generator and judge raises the concern that gains stem from error correlation between annotation and evaluation. We test cross-model configurations on TruthfulQA (full grid in Appendix D.2, Table 9). With LLaMA-3.1 as generator, SAGE $>$ Random $>$ Direct holds for both Qwen3-8B (71.32 vs. 68.11 vs. 65.02%) and Ministral-8B judges (60.34 vs. 58.46 vs. 57.20%). Cross-model SAGE nearly matches same-model: LLaMA-3.1 $\rightarrow$ Qwen3 (71.32%) trails same-model Qwen3 (72.61%) by only 1.29 pp, matching our prediction that reference-based calibration depends on reference usability and the shifts $\delta_1 > 0$, $\delta_0 < 0$, neither requiring a shared model.

Alignment with human evaluation. We validate the model-based ground truth on 817 TruthfulQA samples annotated by four independent raters (Fleiss’ $\kappa=0.552$). SAGE $>$ Random $>$ Direct holds under *every* of seven ground-truth definitions: each rater individually, two majority-

vote thresholds ($\geq 3/4$, $\geq 2/4$), and the 80B judge, with SAGE–Direct gaps of +4.49 to +7.27 pp (table in Appendix D.3).

Sensitivity to k . A sweep over $k \in \{1, 3, 5, 7, 9, 15, 21\}$ on TruthfulQA (Qwen3-8B) shows SAGE beats Direct at every k : even $k=1$ adds +3.50 pp (68.84 vs. 65.34%), AUROC peaks at $k=7-9$ (72.32–72.61%), then declines at $k=15-21$ (diminishing returns under correlated neighbors). Choosing $k=3$ captures most of the gain at $\sim 3\times$ Direct cost (curve in Appendix D.4).

5.4. Summary

SAGE achieves the best performance in 17 out of 24 configurations, with largest gains on HaluEval (+3.8–9.7 points), Flickr30k (+1.4–2.1 points), AG News (+2.6–3.5 points), and CIFAR-10 (+10–21 points). We also analyze retrieval strategies in Appendix C: Unified embeddings match or exceed self-embeddings in 15/18 configurations, enabling practical deployment without generator-specific extraction.

6. Conclusions

We present SAGE, which uses kNN-retrieved semantic neighbors as references for detecting errors in AI-generated annotations. Our theoretical framework reveals that references improve error detection only when semantically matched and correct, a condition kNN retrieval satisfies more frequently than random sampling. Experiments across classification, LLM generation, and VLM captioning validate this, with the largest gains on semantically clustered tasks. SAGE’s effectiveness depends on reference pool quality and the approximate independence assumption, which may be violated with same-model annotations. Future work includes adaptive pool selection, hybrid scoring methods, and cross-model references to reduce systematic biases.

Limitations. (i) *Model-based correctness labels.* For generation tasks, ground truth comes from an external 80B judge; while Appendix D.3 shows the SAGE $>$ Random $>$ Direct ranking holds under every annotator and majority-vote definition, this does not remove dependence on the judge’s calibration. (ii) *Computational cost.* At $k=9$, SAGE issues $\sim 9\times$ the judge calls of Direct; Appendix D.4 shows $k=3$ captures most of the gain at $\sim 3\times$ cost.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning by improving the reliability and evaluation of learning systems through better analysis of annotation quality and model behavior. By focusing on methodological and empirical aspects of data-centric model evaluation, the proposed approach is intended to support more robust and reproducible research practices.

While the techniques introduced in this work may be applied to a wide range of machine learning systems, they do not introduce new capabilities that inherently raise ethical or societal concerns beyond those already present in existing learning frameworks. We believe that improving transparency and reliability in model evaluation can have positive downstream effects by helping practitioners better understand model limitations and reduce unintended errors in real-world deployments. We foresee no negative societal impacts from this work.

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A. Notations

Table 5. Summary of notation used throughout the paper.

Symbol	Description
<i>Input, Output, and Models</i>	
$x \in \mathcal{X}, \hat{a}$	Input instance (question Q , image I , or (I, Q)); generated annotation
$Y, y \in \{0, 1\}$	Correctness indicator and its value (1=correct, 0=incorrect)
G_θ, J_θ, E	Generator, judge (same θ), and embedding encoder
<i>Sets and Indices</i>	
$D, P, \mathcal{I}$	Target set, reference pool, and kNN index
n, i, j, m	#samples, sample index, trial index, method index (1/2/3)
<i>References</i>	
$(x_{\text{ref}}, \hat{a}_{\text{ref}})$	Reference input-annotation pair
$(x_j^{(r)}, \hat{a}_j^{(r)}), (x_j^{(n)}, \hat{a}_j^{(n)})$	j -th random reference (Strategy 2) and kNN reference (Strategy 3)
<i>Scores</i>	
L, k	Maximum score level; number of reference trials
$S^{(m)}, S_{y,j}^{(m)}$	Single-trial score; score for trial j conditioned on class y
$T^{(m)}, T_+, T_-$	Averaged score $\frac{1}{k} \sum_j S_j$; scores for correct/incorrect samples
<i>Usability</i>	
$U_j, M_j, C_j \in \{0, 1\}$	Usability, semantic match, and correctness indicators for reference j
$u_y^{(m)}, N_y^{(m)}, n_1, n_0$	Usability rate $\Pr(U=1 Y=y)$; usability count $\sum_j U_j$; realized counts
<i>Score Decomposition</i>	
$\mu_y^{(m)}, \delta_y^{(m)}, \epsilon_{y,j}^{(m)}$	Baseline mean, mean shift from usable reference, residual noise
$\Delta\mu^{(m)} = \mu_1 - \mu_0$	Mean difference between correct and incorrect classes
<i>Variance and Correlation</i>	
$\sigma_y^2, \sigma_{y,\text{avg}}^2, \sigma_{\text{tot}}$	Per-trial variance, averaged variance, total std $\sqrt{\sigma_{1,\text{avg}}^2 + \sigma_{0,\text{avg}}^2}$
ρ, k_{eff}	Noise correlation; effective sample size $k/(1 + (k-1)\rho)$
<i>Evaluation</i>	
AUROC, $\Phi(\cdot)$, $B_k(n; u)$	Area Under ROC Curve; standard normal CDF; binomial PMF

B. Theoretical Analysis: When and Why Semantic Neighbors Improve Error Detection

This section provides a theoretical framework that answers two central questions: *when* do semantic neighbors improve error detection, and *why*? We develop an analytical model identifying the key factors that govern the effectiveness of neighbor-based reference selection. The analysis is organized as follows:

- Section B.1 introduces the problem setup and notation;
- Section B.2 formalizes reference usability;
- Section B.3 derives the additive surrogate model from mechanistic principles and states the resulting Lemma;
- Section B.4 states the two distributional assumptions;
- Section B.5 develops the AUROC analysis, culminating in a closed-form expression and a single-probit approximation;
- Section B.6 answers the “when” and “why” questions by interpreting the results;
- Section B.7 discusses limitations regarding latent variables and empirical verification.

B.1. Problem Setup and Notation

Task formulation. Let $x \in \mathcal{X}$ denote an input instance: $x = Q$ in LLM settings, or $x = I / (I, Q)$ in VLM settings. A generator G_θ produces a short answer $\hat{a} = G_\theta(x)$. We define the *correctness indicator* (positive class = correct)

$$Y \in \{0, 1\}, \quad Y = 1 \iff \hat{a} \text{ is correct w.r.t. } x. \quad (11)$$

A judge model outputs an *integer* score $S \in \{0, 1, \dots, L\}$ for some fixed L (e.g., $L = 5$), where larger scores indicate

higher answer quality. Given a real-valued score T , we evaluate discrimination using the tie-aware AUROC:

$$\text{AUROC} \triangleq \Pr(T_+ > T_-) + \frac{1}{2} \Pr(T_+ = T_-), \quad (12)$$

where $T_+ \sim T \mid Y = 1$ and $T_- \sim T \mid Y = 0$ are independent.

Three prompting strategies. **(1) Self-score.** The judge scores $(x, \hat{a})$ without any reference:

$$S^{(1)} \triangleq J_\theta(x, \hat{a}), \quad T^{(1)} \triangleq S^{(1)}. \quad (13)$$

Note that for Method (1), both $S^{(1)}$ and $T^{(1)}$ are integers (no averaging).

(2) Random reference (averaged). For each trial $j \in \{1, \dots, k\}$, draw a random reference $(x_j^{(r)}, \hat{a}_j^{(r)})$ from the same generated pool and score:

$$S_j^{(2)} \triangleq J_\theta(x, \hat{a}; x_j^{(r)}, \hat{a}_j^{(r)}), \quad T^{(2)} \triangleq \frac{1}{k} \sum_{j=1}^k S_j^{(2)}. \quad (14)$$

Each $S_j^{(2)} \in \mathbb{Z}$, while $T^{(2)} \in \mathbb{R}$.

(3) kNN reference (averaged). For each trial j , retrieve a neighbor $(x_j^{(n)}, \hat{a}_j^{(n)})$ by cosine similarity under an encoder E and score:

$$S_j^{(3)} \triangleq J_\theta(x, \hat{a}; x_j^{(n)}, \hat{a}_j^{(n)}), \quad T^{(3)} \triangleq \frac{1}{k} \sum_{j=1}^k S_j^{(3)}. \quad (15)$$

For $k \geq 2$, the averaged score $T^{(m)}$ is real-valued and ties in (12) become negligible in practice.

B.2. Reference Usability

A reference improves judging only if it is both semantically transferable and internally correct. We formalize this as follows.

Definition B.1 (Usable Reference). For the j -th reference $(x_{\text{ref},j}, \hat{a}_{\text{ref},j})$ used to judge target $(x, \hat{a})$, define the *usability indicator* as

$$U_j \triangleq \mathbf{1}[M_j = 1 \wedge C_j = 1], \quad (16)$$

where the semantic match indicator $M_j = 1$ iff $x_{\text{ref},j}$ is semantically similar to x , and the correctness indicator $C_j = 1$ iff $\hat{a}_{\text{ref},j}$ is correct for $x_{\text{ref},j}$.

In our setting, reference answers are drawn from the *same generated pool* as $\hat{a}$. Hence C_j is unknown and depends on the generator’s base accuracy and its error correlations within semantic neighborhoods.

B.3. From a Mechanistic Judging Story to the Additive Surrogate Model

This subsection provides a fully explicit derivation of the class-specific additive gain parameters δ_1 and δ_0 used in Lemma B.2. The goal is to make the connection between a mechanistic (reference-calibrated) judging story and an analyzable surrogate model completely transparent.

Notation recap. Recall from Definition B.1 the notion of a *usable reference*. Fix a method $m \in \{2, 3\}$ (random-reference or kNN-reference prompting) and a class label $Y = y$. For a single judging trial j , the prompt includes the target pair $(x, \hat{a})$ and a single reference pair $(x_{\text{ref}}, \hat{a}_{\text{ref}})$ (random or retrieved). The judge outputs an *integer* score $S_{y,j}^{(m)} \in \{0, 1, \dots, L\}$. We defined the *semantic match* indicator $M \in \{0, 1\}$ and the *reference correctness* indicator $C \in \{0, 1\}$, and the *usable reference* indicator U defined in (16). Recall that $U = 1$ means the reference can serve as a trustworthy calibration anchor for judging the target answer; $U = 0$ means the reference is either semantically mismatched or incorrect for itself (and thus potentially misleading).

B.3.1. A GENERAL MECHANISTIC MODEL

We begin with a deliberately general form of “mechanistic” reference-based judging. Let Ω denote the collection of all random factors that affect the judge output (e.g., prompt stochasticity, decoding randomness, context sensitivity). We do not assume any parametric form for these factors.

Mechanistic score generation. We posit that for each trial, the judge output is generated by an (unknown) function of: (i) the target instance and its answer; (ii) the reference instance and its answer; and (iii) random factors:

$$S = \mathcal{J}(x, \hat{a}, x_{\text{ref}}, \hat{a}_{\text{ref}}, \Omega), \quad (17)$$

where $S \in \{0, 1, \dots, L\}$. Equation (17) is *not* yet analyzable for AUROC, but it clarifies that the score distribution changes when the reference changes.

Conditioning on usability. Crucially, all reference effects that matter for our analysis can be organized by the single binary event U in (16). That is, for any class $Y = y$ and any measurable function g , the law of total expectation yields

$$\begin{aligned} \mathbb{E}[g(S) \mid Y = y] &= \mathbb{E}[g(S) \mid Y = y, U = 1] \Pr(U = 1 \mid Y = y) \\ &\quad + \mathbb{E}[g(S) \mid Y = y, U = 0] \Pr(U = 0 \mid Y = y). \end{aligned} \quad (18)$$

We will apply (18) with $g(S) = S$.

B.3.2. DEFINING THE MEAN SHIFT δ EXACTLY

We now introduce δ in the only logically valid way: as a *definition* (a conditional mean difference). Fix $Y = y$. Define the two conditional means

$$\mu_{y,1} \triangleq \mathbb{E}[S \mid Y = y, U = 1], \quad \mu_{y,0} \triangleq \mathbb{E}[S \mid Y = y, U = 0]. \quad (19)$$

These expectations are taken over all randomness in the reference sampling/retrieval and the judge process Ω .

Definition of the additive gain. Define the *usable-reference mean shift* for class y as

$$\delta_y \triangleq \mu_{y,1} - \mu_{y,0} = \mathbb{E}[S \mid Y = y, U = 1] - \mathbb{E}[S \mid Y = y, U = 0]. \quad (20)$$

Equation (20) is an identity: it introduces δ_y as a name for a measurable quantity. No approximation has been made.

Expected signs of δ_1 and δ_0 . With our convention that larger scores indicate higher quality and $Y = 1$ denotes correct answers, we expect the class-specific effects to have opposite signs. Intuitively, a usable reference (semantically matched and correct) should help the judge *validate* correct answers (raising their scores, so $\delta_1 > 0$) and *detect* incorrect answers by contrast (lowering their scores, so $\delta_0 < 0$). We defer discussion of how the signs of δ_y affect discrimination to Section B.6.

B.3.3. DERIVING THE ADDITIVE DECOMPOSITION

We next show how the additive form in Lemma B.2 follows from basic conditional expectation identities.

Step 1: Express the conditional mean as an affine function of U . Because U is binary, for any fixed $Y = y$, the conditional mean $\mathbb{E}[S \mid Y = y, U]$ can be written exactly as

$$\begin{aligned} \mathbb{E}[S \mid Y = y, U] &= \mu_{y,0} \cdot \mathbf{1}[U = 0] + \mu_{y,1} \cdot \mathbf{1}[U = 1] \\ &= \mu_{y,0} + (\mu_{y,1} - \mu_{y,0})U \\ &= \mu_{y,0} + \delta_y U, \end{aligned} \quad (21)$$

where we used $\mathbf{1}[U = 1] = U$ and $\mathbf{1}[U = 0] = 1 - U$, and the definition of δ_y in (20). Equation (21) is exact.

Step 2: Define the residual noise term. Define the residual (zero-mean) noise by subtracting the conditional mean:

$$\epsilon \triangleq S - \mathbb{E}[S \mid Y = y, U]. \quad (22)$$

Taking conditional expectation of (22) with respect to $(Y = y, U)$ gives

$$\mathbb{E}[\epsilon \mid Y = y, U] = \mathbb{E}[S \mid Y = y, U] - \mathbb{E}[S \mid Y = y, U] = 0. \quad (23)$$

Step 3: Substitute the affine conditional mean into the score. Combining (21) and (22), we obtain the exact decomposition

$$S = \mu_{y,0} + \delta_y U + \epsilon, \quad \text{with } \mathbb{E}[\epsilon \mid Y = y, U] = 0. \quad (24)$$

This shows that an *additive* representation always exists whenever we summarize reference effects by a binary usability indicator U . No approximation is required for (24); it is a consequence of the definition of δ_y and the residual ϵ .

Matching notation to Lemma B.2. For method m and class y , we identify

$$\mu_y^{(m)} \triangleq \mu_{y,0}, \quad \delta_y^{(m)} \triangleq \delta_y, \quad \epsilon_{y,j}^{(m)} \triangleq \epsilon, \quad (25)$$

yielding the per-trial form

$$S_{y,j}^{(m)} = \mu_y^{(m)} + \delta_y^{(m)} U_{y,j}^{(m)} + \epsilon_{y,j}^{(m)}, \quad \mathbb{E}[\epsilon_{y,j}^{(m)} \mid Y = y, U_{y,j}^{(m)}] = 0, \quad (26)$$

which is exactly the statement of Lemma B.2, now derived with explicit definitions.

B.3.4. EXTENSION TO k -TRIAL AVERAGING

To move from single trials to the averaged scoring protocol used in practice, we now aggregate over k reference trials. For the j -th trial, let $U_{y,j}^{(m)}$ denote usability and $S_{y,j}^{(m)}$ the corresponding score. Averaging (26) over $j = 1, \dots, k$ yields

$$\begin{aligned} T_y^{(m)} &= \frac{1}{k} \sum_{j=1}^k S_{y,j}^{(m)} \\ &= \mu_y^{(m)} + \delta_y^{(m)} \cdot \frac{1}{k} \sum_{j=1}^k U_{y,j}^{(m)} + \frac{1}{k} \sum_{j=1}^k \epsilon_{y,j}^{(m)}. \end{aligned} \quad (27)$$

Defining $N_y^{(m)} \triangleq \sum_{j=1}^k U_{y,j}^{(m)}$ (the number of usable references; see (34)) and $\bar{\epsilon}_y^{(m)} \triangleq \frac{1}{k} \sum_{j=1}^k \epsilon_{y,j}^{(m)}$ gives

$$T_y^{(m)} = \mu_y^{(m)} + \delta_y^{(m)} \frac{N_y^{(m)}}{k} + \bar{\epsilon}_y^{(m)}, \quad (28)$$

which matches the averaged form in (30).

B.3.5. SUMMARY: ADDITIVE SCORE DECOMPOSITION

The derivation above establishes the following result.

Lemma B.2 (Additive score decomposition). *For method $m \in \{2, 3\}$ and class $Y = y$, each per-trial integer score admits the exact decomposition*

$$S_{y,j}^{(m)} = \mu_y^{(m)} + \delta_y^{(m)} U_{y,j}^{(m)} + \epsilon_{y,j}^{(m)}, \quad (29)$$

where $U_{y,j}^{(m)}$ denotes the usability indicator (Definition B.1) for the j -th reference under method m when the target has class $Y = y$, $\mu_y^{(m)}$ is a baseline mean score, $\delta_y^{(m)}$ is the class-specific mean shift contributed by a usable reference, and $\epsilon_{y,j}^{(m)}$ is zero-mean judge noise. Averaging over k trials yields

$$T_y^{(m)} = \mu_y^{(m)} + \delta_y^{(m)} \frac{1}{k} \sum_{j=1}^k U_{y,j}^{(m)} + \bar{\epsilon}_y^{(m)}, \quad (30)$$

where $\bar{\epsilon}_y^{(m)} = \frac{1}{k} \sum_{j=1}^k \epsilon_{y,j}^{(m)}$.

B.4. Distributional Assumptions

To enable tractable AUROC analysis, we introduce two distributional assumptions. Assumption B.4 approximates the averaged score distribution as Gaussian; Assumption B.4 models the count of usable references as Binomial.

Assumption B (Normal approximation for averaged scores). When the averaged residual noise $\bar{\epsilon}_y^{(m)}$ aggregates many weakly-dependent components and has light tails, we approximate its conditional distribution by a Gaussian:

$$T^{(m)} \mid (Y = y, N_y^{(m)} = n) \approx \mathcal{N}\left(\mu_y^{(m)} + \delta_y^{(m)} \frac{n}{k}, \sigma_{y,\text{avg}}^{2,(m)}\right), \quad (31)$$

where $N_y^{(m)} = \sum_{j=1}^k U_{y,j}^{(m)}$ is the number of usable references, and the variance $\sigma_{y,\text{avg}}^{2,(m)}$ accounts for correlated noise across trials (derived in Section B.5.2).

Conditions for validity. The Gaussian approximation is justified under the following conditions:

- (B1) **Moderate k :** The number of reference trials k should be at least 5. Intuitively, five averaged bounded scores provide enough “granularity” for the CLT to smooth out discrete effects; with $k < 5$, the averaged score takes only a few distinct values, making the continuous Gaussian model a poor fit.
- (B2) **Bounded scores:** Per-trial scores $S_{y,j}^{(m)} \in \{0, 1, \dots, L\}$ are bounded integers, ensuring finite variance and sub-Gaussian tails for the residual noise $\epsilon_{y,j}^{(m)}$. This boundedness guarantees that rare, extreme outliers cannot dominate the average.
- (B3) **Moderate correlation:** The pairwise noise correlation $\rho_y^{(m)}$ should satisfy $\rho_y^{(m)} < 1$. When $\rho \rightarrow 1$, all trials carry essentially the same information (perfect correlation means no new signal from additional references), so $k_{\text{eff}} \rightarrow 1$ and averaging provides no variance reduction.

Failure modes. The approximation may degrade under the following scenarios:

- **Very small k (1–2):** The CLT approximation is poor, and discrete score effects dominate. The averaged score takes only finitely many values, making the continuous Gaussian model inaccurate.
- **High correlation ($\rho \rightarrow 1$):** When trial noises are nearly perfectly correlated (e.g., strong same-model bias), the effective sample size approaches 1, and the benefits of averaging vanish.
- **Multimodal score distributions:** If the judge produces bimodal scores (e.g., scores clustering at 1 and 5 with nothing in between), the Gaussian approximation underestimates tail probabilities.

Assumption C (Approximately independent usability). For fixed m and y , the usability indicators $\{U_{y,j}^{(m)}\}_{j=1}^k$ are approximately independent with

$$u_y^{(m)} \triangleq \Pr(U_{y,j}^{(m)} = 1 \mid Y = y). \quad (32)$$

Then $N_y^{(m)} \sim \text{Binomial}(k, u_y^{(m)})$. This can be relaxed to Beta–Binomial to model same-model error coupling.

Conditions for validity. The independence approximation is justified under the following conditions:

- (C1) **Random references ($m = 2$):** For random sampling, references are drawn i.i.d. from the reference pool, so usability indicators are exactly independent.
- (C2) **Conditional independence of errors ($m = 3$):** For kNN retrieval, neighbors are semantically similar by construction. However, usability independence holds approximately under the assumption that annotation errors are *conditionally independent given the ground-truth label*, a standard assumption in the noisy label literature (Zhu et al., 2021b). That is, while neighbors share similar semantic content, whether each neighbor’s annotation is correct depends primarily on the ground-truth correctness of the target, with residual variation being approximately independent across neighbors.
- (C3) **Low same-model error correlation:** Annotation errors should not be strongly correlated across semantically similar instances. This is more likely when annotations are generated with sampling randomness, or when the generator does not have systematic blind spots.

Failure modes. The independence assumption may be violated in the following scenarios:

- **Same-model error coupling:** When the same generator model produces all annotations, errors often cluster semantically. Incorrect answers to similar questions may share the same misconception (e.g., all questions about a specific topic are answered incorrectly due to a knowledge gap). This induces positive correlation among usability indicators for kNN references, inflating the variance of $N_y^{(m)}$ beyond the Binomial model.
- **Tight kNN clustering:** When the k nearest neighbors are all very similar to each other (low diversity in embedding space), their correctness indicators C_j become highly correlated. If one neighbor is incorrect, others are likely incorrect as well.
- **Systematic generator errors:** If the generator has consistent failure modes on certain question types (e.g., mathematical reasoning, temporal questions), neighbors within those categories will have correlated errors.

Relaxation via Beta–Binomial. When independence is violated, the count $N_y^{(m)}$ exhibits *overdispersion*, i.e., variance larger than the Binomial model predicts. The Beta–Binomial distribution captures this by introducing a correlation parameter $\rho_U \in [0, 1]$:

$$\text{Var}(N_y^{(m)}) = k u_y^{(m)} (1 - u_y^{(m)}) \cdot \underbrace{(1 + (k-1)\rho_U)}_{\kappa \geq 1}. \quad (33)$$

When $\rho_U = 0$, this reduces to the Binomial variance; when $\rho_U > 0$, the overdispersion factor $\kappa > 1$ increases variance. In practice, when annotations are generated by a single model, setting $\rho_U \in [0.05, 0.2]$ is a reasonable starting point based on typical same-model error correlations observed in LLM/VLM settings. The single-probit approximation in Corollary B.6 remains valid under Beta–Binomial when the variance contribution from (66) remains small relative to score noise.

B.5. AUROC Analysis

This section develops the AUROC analysis in five steps: (1) an exact mixture decomposition (Section B.5.1); (2) variance derivation for correlated noise (Section B.5.2); (3) pairwise comparison under Gaussianity (Section B.5.3); (4) a closed-form AUROC under Lemma B.2 and Assumptions B.4–B.4 (Section B.5.4); and (5) a single-probit approximation (Section B.5.5).

B.5.1. EXACT MIXTURE DECOMPOSITION

Define the number of usable references among k trials:

$$N_y^{(m)} \triangleq \sum_{j=1}^k U_{y,j}^{(m)}. \quad (34)$$

Let $\mathcal{L}_{y,n}^{(m)}$ denote the conditional distribution of $T^{(m)}$ given $(Y = y, N_y^{(m)} = n)$, with CDF $F_{y,n}^{(m)}$.

Lemma B.3 (Exact mixture decomposition of AUROC). *For $m \in \{2, 3\}$,*

$$\text{AUROC}^{(m)} = \sum_{n_1=0}^k \sum_{n_0=0}^k \Pr(N_1^{(m)} = n_1 \mid Y = 1) \Pr(N_0^{(m)} = n_0 \mid Y = 0) \Gamma_{n_1, n_0}^{(m)}, \quad (35)$$

where

$$\Gamma_{n_1, n_0}^{(m)} = \Pr(T_{1, n_1}^{(m)} > T_{0, n_0}^{(m)}) + \frac{1}{2} \Pr(T_{1, n_1}^{(m)} = T_{0, n_0}^{(m)}), \quad (36)$$

and $T_{y,n}^{(m)} \sim \mathcal{L}_{y,n}^{(m)}$ are independent draws.

Proof. By definition of AUROC in (12),

$$\text{AUROC}^{(m)} = \Pr(T_+^{(m)} > T_-^{(m)}) + \frac{1}{2} \Pr(T_+^{(m)} = T_-^{(m)}),$$

where $T_+^{(m)} \sim T^{(m)} \mid Y = 1$ and $T_-^{(m)} \sim T^{(m)} \mid Y = 0$ are independent. Since the distribution of $T^{(m)}$ given $Y = y$

depends on $N_y^{(m)}$, we apply the law of total probability by conditioning on $(N_1^{(m)}, N_0^{(m)})$:

$$\begin{aligned} \text{AUROC}^{(m)} &= \sum_{n_1=0}^k \sum_{n_0=0}^k \Pr(N_1^{(m)} = n_1 \mid Y = 1) \Pr(N_0^{(m)} = n_0 \mid Y = 0) \\ &\quad \cdot \left[\Pr(T_{1,n_1}^{(m)} > T_{0,n_0}^{(m)}) + \frac{1}{2} \Pr(T_{1,n_1}^{(m)} = T_{0,n_0}^{(m)}) \right], \end{aligned}$$

where we used independence of $N_1^{(m)}$ and $N_0^{(m)}$ (they correspond to different instances). Defining $\Gamma_{n_1, n_0}^{(m)}$ as the bracketed term completes the proof. $\square$

Lemma B.3 is exact and holds for arbitrary score distributions, including discrete per-trial scores and correlated references.

B.5.2. VARIANCE OF AVERAGED SCORES WITH CORRELATED NOISE

In this subsection, we derive the variance of the averaged residual noise $\bar{\epsilon}_y^{(m)} = \frac{1}{k} \sum_{j=1}^k \epsilon_{y,j}^{(m)}$ under a simple but expressive correlation model.

Noise model. Fix a method m and class $Y = y$. Assume the per-trial residual noises $\{\epsilon_{y,j}^{(m)}\}_{j=1}^k$ satisfy:

$$\mathbb{E}[\epsilon_{y,j}^{(m)}] = 0, \quad (37)$$

$$\text{Var}(\epsilon_{y,j}^{(m)}) = \sigma_y^{2,(m)}, \quad (38)$$

$$\text{Cov}(\epsilon_{y,i}^{(m)}, \epsilon_{y,j}^{(m)}) = \rho_y^{(m)} \sigma_y^{2,(m)}, \quad i \neq j, \quad (39)$$

where $\rho_y^{(m)} \in [0, 1)$ denotes the pairwise correlation coefficient. This equicorrelation structure captures shared prompt effects, same-model biases, and other common sources of dependence across trials.

Variance of the averaged noise. By definition,

$$\bar{\epsilon}_y^{(m)} = \frac{1}{k} \sum_{j=1}^k \epsilon_{y,j}^{(m)}. \quad (40)$$

Using bilinearity of covariance, we compute

$$\begin{aligned} \text{Var}(\bar{\epsilon}_y^{(m)}) &= \text{Var}\left(\frac{1}{k} \sum_{j=1}^k \epsilon_{y,j}^{(m)}\right) \\ &= \frac{1}{k^2} \text{Var}\left(\sum_{j=1}^k \epsilon_{y,j}^{(m)}\right) \\ &= \frac{1}{k^2} \left(\sum_{j=1}^k \text{Var}(\epsilon_{y,j}^{(m)}) + 2 \sum_{1 \leq i < j \leq k} \text{Cov}(\epsilon_{y,i}^{(m)}, \epsilon_{y,j}^{(m)}) \right). \end{aligned} \quad (41)$$

Substituting the noise moments. Substituting (38) and (39) into (41), we obtain

$$\begin{aligned} \text{Var}(\bar{\epsilon}_y^{(m)}) &= \frac{1}{k^2} \left(k \sigma_y^{2,(m)} + 2 \binom{k}{2} \rho_y^{(m)} \sigma_y^{2,(m)} \right) \\ &= \frac{1}{k^2} \left(k \sigma_y^{2,(m)} + k(k-1) \rho_y^{(m)} \sigma_y^{2,(m)} \right) \\ &= \sigma_y^{2,(m)} \left(\frac{1}{k} + \frac{k-1}{k} \rho_y^{(m)} \right). \end{aligned} \quad (42)$$

Effective sample size. It is convenient to express (42) in the form

$$\text{Var}(\bar{\epsilon}_y^{(m)}) = \frac{\sigma_y^{2,(m)}}{k_{\text{eff},y}^{(m)}}, \quad (43)$$

where the *effective sample size* is defined as

$$k_{\text{eff},y}^{(m)} \triangleq \frac{k}{1 + (k-1)\rho_y^{(m)}}. \quad (44)$$

Intuitively, when trials are independent ($\rho = 0$), all k trials are fully informative, so $k_{\text{eff}} = k$; as correlation increases, some trials become redundant (they carry overlapping information), reducing the effective number of independent measurements. When $\rho_y^{(m)} = 0$, $k_{\text{eff},y}^{(m)} = k$ recovers the i.i.d. case. As $\rho_y^{(m)}$ increases, $k_{\text{eff},y}^{(m)}$ decreases, reflecting diminishing variance reduction due to correlated trials.

Connection to Assumption B.4. The variance $\sigma_{y,\text{avg}}^{2,(m)}$ appearing in Assumption B.4 is precisely $\text{Var}(\bar{\epsilon}_y^{(m)})$ derived above:

$$\sigma_{y,\text{avg}}^{2,(m)} = \frac{\sigma_y^{2,(m)}}{k_{\text{eff},y}^{(m)}} = \frac{\sigma_y^{2,(m)}(1 + (k-1)\rho_y^{(m)})}{k}. \quad (45)$$

B.5.3. PAIRWISE COMPARISON UNDER GAUSSIANITY

The following lemma computes the pairwise comparison probability needed for the AUROC mixture formula.

Lemma B.4 (Pairwise comparison under conditional Gaussianity). *Assume the conditional Gaussian model in (31), i.e., for $y \in \{0, 1\}$ and $n \in \{0, \dots, k\}$,*

$$T_{y,n}^{(m)} \sim \mathcal{N}\left(\mu_y^{(m)} + \delta_y^{(m)} \frac{n}{k}, \sigma_{y,\text{avg}}^{2,(m)}\right), \quad (46)$$

and $T_{1,n_1}^{(m)}$ and $T_{0,n_0}^{(m)}$ are independent.

Then the probability $\Pr(T_{1,n_1}^{(m)} > T_{0,n_0}^{(m)})$ admits the closed form

$$\Pr\left(T_{1,n_1}^{(m)} > T_{0,n_0}^{(m)}\right) = \Phi\left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)} n_1/k - \delta_0^{(m)} n_0/k}{\sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}}\right), \quad (47)$$

where $\Delta\mu^{(m)} = \mu_1^{(m)} - \mu_0^{(m)}$ and Φ denotes the standard normal cumulative distribution function.

Proof. We provide a step-by-step derivation.

Step 1: Define the difference random variable. Define

$$D \triangleq T_{1,n_1}^{(m)} - T_{0,n_0}^{(m)}. \quad (48)$$

Then

$$\Pr(T_{1,n_1}^{(m)} > T_{0,n_0}^{(m)}) = \Pr(D > 0). \quad (49)$$

Step 2: Distribution of the difference. Since $T_{1,n_1}^{(m)}$ and $T_{0,n_0}^{(m)}$ are independent Gaussian random variables, their difference D is also Gaussian. Its mean is

$$\begin{aligned} \mathbb{E}[D] &= \mathbb{E}[T_{1,n_1}^{(m)}] - \mathbb{E}[T_{0,n_0}^{(m)}] \\ &= \left(\mu_1^{(m)} + \delta_1^{(m)} \frac{n_1}{k}\right) - \left(\mu_0^{(m)} + \delta_0^{(m)} \frac{n_0}{k}\right) \\ &= \Delta\mu^{(m)} + \delta_1^{(m)} \frac{n_1}{k} - \delta_0^{(m)} \frac{n_0}{k}, \end{aligned} \quad (50)$$

and its variance is

$$\begin{aligned}\text{Var}(D) &= \text{Var}(T_{1,n_1}^{(m)}) + \text{Var}(T_{0,n_0}^{(m)}) \\ &= \sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)},\end{aligned}\tag{51}$$

where we used independence to drop the covariance term.

Thus,

$$D \sim \mathcal{N}\left(\Delta\mu^{(m)} + \delta_1^{(m)} \frac{n_1}{k} - \delta_0^{(m)} \frac{n_0}{k}, \sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}\right).\tag{52}$$

Step 3: Standardization. Define the standardized variable

$$Z \triangleq \frac{D - \mathbb{E}[D]}{\sqrt{\text{Var}(D)}} \sim \mathcal{N}(0, 1).\tag{53}$$

Then

$$\begin{aligned}\Pr(D > 0) &= \Pr\left(Z > \frac{0 - \mathbb{E}[D]}{\sqrt{\text{Var}(D)}}\right) \\ &= 1 - \Phi\left(\frac{-\mathbb{E}[D]}{\sqrt{\text{Var}(D)}}\right) \\ &= \Phi\left(\frac{\mathbb{E}[D]}{\sqrt{\text{Var}(D)}}\right),\end{aligned}\tag{54}$$

where the last equality follows from the symmetry $\Phi(-x) = 1 - \Phi(x)$.

Step 4: Definition of Φ . The standard normal cumulative distribution function is defined as

$$\Phi(t) \triangleq \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du.\tag{55}$$

Step 5: Substitution. Substituting (50) and (51) into (54) yields the stated result. $\square$

B.5.4. CLOSED-FORM AUROC

We now combine all preceding results to derive a closed-form AUROC expression. The theorem below relies on Lemma B.3 (exact mixture decomposition), Assumption B.4 (conditional Gaussianity, which enables Lemma B.4), and Assumption B.4 (Binomial distribution of usable-reference counts). Note that Lemma B.2 underlies Assumption B.4 by providing the conditional mean structure.

Theorem B.5 (Closed-form mixture AUROC for k -reference averaging). *Under Lemma B.2 and Assumptions B.4–B.4, for $m \in \{2, 3\}$,*

$$\begin{aligned}\text{AUROC}^{(m)} &= \sum_{n_1=0}^k \sum_{n_0=0}^k \binom{k}{n_1} (u_1^{(m)})^{n_1} (1 - u_1^{(m)})^{k-n_1} \binom{k}{n_0} (u_0^{(m)})^{n_0} (1 - u_0^{(m)})^{k-n_0} \\ &\quad \cdot \Phi\left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)} n_1/k - \delta_0^{(m)} n_0/k}{\sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}}\right),\end{aligned}\tag{56}$$

where $\Delta\mu^{(m)} = \mu_1^{(m)} - \mu_0^{(m)}$ and $\Phi(\cdot)$ is the standard normal CDF.

Proof. We expand all steps explicitly.

Step 1: Start from the exact AUROC mixture decomposition. Lemma B.3 states that, for $m \in \{2, 3\}$,

$$\text{AUROC}^{(m)} = \sum_{n_1=0}^k \sum_{n_0=0}^k \Pr(N_1^{(m)} = n_1 \mid Y = 1) \Pr(N_0^{(m)} = n_0 \mid Y = 0) \Gamma_{n_1, n_0}^{(m)}, \quad (57)$$

where

$$\Gamma_{n_1, n_0}^{(m)} = \Pr\left(T_{1, n_1}^{(m)} > T_{0, n_0}^{(m)}\right) + \frac{1}{2} \Pr\left(T_{1, n_1}^{(m)} = T_{0, n_0}^{(m)}\right). \quad (58)$$

Step 2: Evaluate the count probabilities using Assumption B.4. Assumption B.4 states that for class $Y = y$, $N_y^{(m)} = \sum_{j=1}^k U_{y,j}^{(m)}$ follows a Binomial distribution with parameter $u_y^{(m)} = \Pr(U_{y,j}^{(m)} = 1 \mid Y = y)$. Hence for each $n \in \{0, \dots, k\}$,

$$\Pr(N_y^{(m)} = n \mid Y = y) = \binom{k}{n} (u_y^{(m)})^n (1 - u_y^{(m)})^{k-n}. \quad (59)$$

Substituting (59) into (57) yields

$$\begin{aligned} \text{AUROC}^{(m)} &= \sum_{n_1=0}^k \sum_{n_0=0}^k \binom{k}{n_1} (u_1^{(m)})^{n_1} (1 - u_1^{(m)})^{k-n_1} \binom{k}{n_0} (u_0^{(m)})^{n_0} (1 - u_0^{(m)})^{k-n_0} \\ &\quad \cdot \Gamma_{n_1, n_0}^{(m)}. \end{aligned} \quad (60)$$

Step 3: Evaluate $\Gamma_{n_1, n_0}^{(m)}$ under Assumption B.4. Assumption B.4 states that conditional on $(Y = y, N_y^{(m)} = n)$,

$$T_{y,n}^{(m)} \sim \mathcal{N}\left(\mu_y^{(m)} + \delta_y^{(m)} \frac{n}{k}, \sigma_{y,\text{avg}}^{2,(m)}\right), \quad (61)$$

and the draws $T_{1,n_1}^{(m)}$ and $T_{0,n_0}^{(m)}$ are independent. Since Gaussian distributions are continuous, we have

$$\Pr\left(T_{1,n_1}^{(m)} = T_{0,n_0}^{(m)}\right) = 0, \quad (62)$$

and therefore

$$\Gamma_{n_1, n_0}^{(m)} = \Pr\left(T_{1,n_1}^{(m)} > T_{0,n_0}^{(m)}\right). \quad (63)$$

Step 4: Use the Gaussian pairwise comparison lemma. By Lemma B.4,

$$\Pr\left(T_{1,n_1}^{(m)} > T_{0,n_0}^{(m)}\right) = \Phi\left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)} n_1/k - \delta_0^{(m)} n_0/k}{\sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}}\right). \quad (64)$$

Combining (60), (63), and (64) gives exactly (56). $\square$

Method (1) as a degenerate case. Method (1) (self-score without reference) can be viewed as a degenerate instance of the additive framework. Specifically, there is no reference usability variable ($U \equiv 0$), no averaging ($k = 1$), and thus no reference-induced shift. Formally, this corresponds to setting

$$u_y^{(1)} = 0, \quad \delta^{(1)} = 0,$$

so that the score reduces to

$$T^{(1)} = \mu_y^{(1)} + \epsilon_y^{(1)}.$$

Its AUROC is therefore determined solely by the intrinsic separability $\Delta\mu^{(1)}$ and the noise variance of the self-judging score.

B.5.5. SINGLE-PROBIT APPROXIMATION

Under Assumption B.4 (or its Beta–Binomial relaxation),

$$\mathbb{E}[N_y^{(m)}] = k u_y^{(m)}, \quad \text{Var}(N_y^{(m)}) = k u_y^{(m)} (1 - u_y^{(m)}) \cdot \kappa_y^{(m)}, \quad (65)$$

where $\kappa_y^{(m)} \geq 1$ is the overdispersion factor. Under exact Assumption B.4, $\kappa_y^{(m)} = 1$; under Beta–Binomial, $\kappa_y^{(m)} = 1 + (k - 1)\rho_U$ for some usability correlation $\rho_U \in [0, 1]$.

The random shift term $\delta_1^{(m)} N_1^{(m)} / k - \delta_0^{(m)} N_0^{(m)} / k$ has variance

$$\frac{(\delta_1^{(m)})^2 u_1^{(m)} (1 - u_1^{(m)}) \kappa_1^{(m)}}{k} + \frac{(\delta_0^{(m)})^2 u_0^{(m)} (1 - u_0^{(m)}) \kappa_0^{(m)}}{k}. \quad (66)$$

Approximation rationale. Equation (56) involves an expectation of $\Phi(\cdot)$ over the random counts $(N_1^{(m)}, N_0^{(m)})$. When the variance of the random shift in (66) is small compared to the intrinsic score noise, the argument of $\Phi(\cdot)$ concentrates around its mean. In this regime, we approximate the mixture by evaluating $\Phi(\cdot)$ at the mean shift, leading to a single-probit expression.

Corollary B.6 (Single-probit AUROC approximation). *If (66) is small relative to $\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}$, then*

$$\text{AUROC}^{(m)} \approx \Phi \left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)} u_1^{(m)} - \delta_0^{(m)} u_0^{(m)}}{\sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}} \right). \quad (67)$$

Proof. Define the function

$$f(n_1, n_0) \triangleq \Phi \left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)} n_1 / k - \delta_0^{(m)} n_0 / k}{\sigma_{\text{tot}}} \right),$$

where $\sigma_{\text{tot}} \triangleq \sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}$ for brevity. By Theorem B.5,

$$\text{AUROC}^{(m)} = \mathbb{E} \left[f(N_1^{(m)}, N_0^{(m)}) \right].$$

Step 1: Taylor expansion. Let $\bar{n}_y \triangleq \mathbb{E}[N_y^{(m)}] = k u_y^{(m)}$ denote the mean count for class y . Expand f around $(\bar{n}_1, \bar{n}_0)$ to second order:

$$\begin{aligned} f(n_1, n_0) &= f(\bar{n}_1, \bar{n}_0) + \frac{\partial f}{\partial n_1} \Big|_{(\bar{n}_1, \bar{n}_0)} (n_1 - \bar{n}_1) + \frac{\partial f}{\partial n_0} \Big|_{(\bar{n}_1, \bar{n}_0)} (n_0 - \bar{n}_0) \\ &\quad + \frac{1}{2} \frac{\partial^2 f}{\partial n_1^2} \Big|_{(\bar{n}_1, \bar{n}_0)} (n_1 - \bar{n}_1)^2 + \frac{1}{2} \frac{\partial^2 f}{\partial n_0^2} \Big|_{(\bar{n}_1, \bar{n}_0)} (n_0 - \bar{n}_0)^2 \\ &\quad + \frac{\partial^2 f}{\partial n_1 \partial n_0} \Big|_{(\bar{n}_1, \bar{n}_0)} (n_1 - \bar{n}_1)(n_0 - \bar{n}_0) + O((n - \bar{n})^3). \end{aligned}$$

Step 2: Compute derivatives. Let $z(n_1, n_0) \triangleq \frac{\Delta\mu^{(m)} + \delta_1^{(m)} n_1 / k - \delta_0^{(m)} n_0 / k}{\sigma_{\text{tot}}}$ so that $f = \Phi(z)$. Then $\frac{\partial f}{\partial n_1} = \phi(z) \frac{\delta_1^{(m)}}{k \sigma_{\text{tot}}}$ and $\frac{\partial f}{\partial n_0} = -\phi(z) \frac{\delta_0^{(m)}}{k \sigma_{\text{tot}}}$, where ϕ is the standard normal PDF. The second derivatives involve $\phi'(z) = -z\phi(z)$ and are $O(\max\{(\delta_1^{(m)})^2, (\delta_0^{(m)})^2\} / (k \sigma_{\text{tot}})^2)$.

Step 3: Take expectations. Taking expectations and using $\mathbb{E}[N_y^{(m)} - \bar{n}_y] = 0$:

$$\mathbb{E}[f(N_1^{(m)}, N_0^{(m)})] = f(\bar{n}_1, \bar{n}_0) + O \left(\frac{\text{variance terms from (66)}}{(k \sigma_{\text{tot}})^2} \right).$$

The remainder is precisely controlled by (66) relative to σ_{tot}^2 .

Step 4: Substitute the mean. When the remainder is negligible,

$$\begin{aligned} \text{AUROC}^{(m)} &\approx f(\bar{n}_1, \bar{n}_0) \\ &= \Phi \left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)}(ku_1^{(m)})/k - \delta_0^{(m)}(ku_0^{(m)})/k}{\sigma_{\text{tot}}} \right) \\ &= \Phi \left(\frac{\Delta\mu^{(m)} + \delta_1^{(m)}u_1^{(m)} - \delta_0^{(m)}u_0^{(m)}}{\sigma_{\text{tot}}} \right), \end{aligned}$$

which is exactly (67). $\square$

B.6. When and Why Semantic Neighbors Help

This section provides a detailed analysis extending the insights from main text Section 4.4. Under the single-probit approximation (67), the AUROC of all three prompting strategies can be expressed in a unified form:

$$\text{AUROC}^{(m)} \approx \Phi \left(\frac{\underbrace{\Delta\mu^{(m)}}_{\text{intrinsic separability}} + \underbrace{\delta_1^{(m)}u_1^{(m)} - \delta_0^{(m)}u_0^{(m)}}_{\text{reference-induced mean shift}}}{\underbrace{\sqrt{\sigma_{1,\text{avg}}^{2,(m)} + \sigma_{0,\text{avg}}^{2,(m)}}}_{\text{effective noise after averaging}}} \right). \quad (68)$$

This decomposition makes explicit that detection performance is governed by *three conceptually distinct factors*: (i) the judge’s intrinsic ability to separate correct from incorrect answers ($\Delta\mu^{(m)}$); (ii) the additional mean separation induced by usable references ($\delta_1^{(m)}u_1^{(m)} - \delta_0^{(m)}u_0^{(m)}$); and (iii) the effective noise level after averaging (controlled by $k_{\text{eff}}^{(m)}$). We now analyze each method’s mechanism and effectiveness through these three factors.

Method (1): Self-score (no reference). Method (1) corresponds to a degenerate case with no reference usage. Since no reference is provided, the usability indicators and δ terms are not applicable ($k = 1$, no U). Consequently, AUROC depends solely on the intrinsic separability $\Delta\mu^{(1)}$ and the raw judging noise. There is neither variance reduction through averaging nor any reference-induced calibration. Method (1) is therefore robust but limited: its performance is capped by the judge’s standalone reliability.

Method (2): Random references. Method (2) introduces multiple reference trials ($k > 1$), yielding variance reduction through an effective sample size $k_{\text{eff}}^{(2)} > 1$. Because random references are unlikely to be semantically aligned with the query, both usable-reference rates are small and approximately equal: $u_1^{(2)} \approx u_0^{(2)} \approx (\text{small})$. As a result, the reference-induced mean shift $\delta_1^{(2)}u_1^{(2)} - \delta_0^{(2)}u_0^{(2)}$ is small in magnitude. Thus, Method (2) improves AUROC primarily by reducing noise (factor iii), not by increasing class separation (factor ii). It serves as a variance-reduction baseline rather than a semantic calibration mechanism.

Method (3): Semantic neighbors. Method (3) leverages semantic retrieval to select neighbors that are more likely to provide useful reference-based calibration. Correct answers tend to have semantically similar neighbors that are also correct, whereas incorrect answers may retrieve neighbors that are themselves incorrect (due to correlated errors), leading to $u_1^{(3)} > u_0^{(3)}$. While averaging across trials reduces noise (factor iii), Method (3) can *additionally* increase mean separation (factor ii) through the term $\delta_1^{(3)}u_1^{(3)} - \delta_0^{(3)}u_0^{(3)}$, and therefore can outperform Methods (1) and (2) when semantic matching is effective and same-model correlations remain moderate.

Why semantic neighbors help: the role of δ signs. The signs of $\delta_1^{(m)}$ and $\delta_0^{(m)}$ reveal the *mechanism* through which semantic neighbors improve error detection. Under the expected sign conditions ($\delta_1^{(m)} > 0$, $\delta_0^{(m)} < 0$):

- The term $\delta_1^{(m)}u_1^{(m)} > 0$ indicates that usable references *raise* scores for correct answers;
- The term $-\delta_0^{(m)}u_0^{(m)} > 0$ indicates that usable references *lower* scores for incorrect answers.

In this case, *both* terms contribute positively to the numerator in (68), improving AUROC through two complementary mechanisms. This is precisely *why* semantic neighbors help: semantically matched, correct references provide calibration anchors that help the judge validate correct answers and expose incorrect ones.

If the signs differ from this expectation, the interpretation changes accordingly. For instance, if $\delta_0^{(m)} > 0$ (usable references also raise scores for incorrect answers), then the benefit from the $\delta_1^{(m)} u_1^{(m)}$ term would be partially offset, and the net effect depends on the relative magnitudes.

When semantic neighbors may not help. The advantage of Method (3) is not unconditional:

- **Weak semantic-correctness correlation:** If semantic similarity does not correlate with reference correctness (both $u_1^{(3)}$ and $u_0^{(3)}$ are small), the mean-shift term becomes negligible and Method (3) degenerates to Method (2).
- **Strong same-model coupling:** If same-model correlations inflate noise across trials, $k_{\text{eff}}^{(3)}$ approaches 1, and Method (3) further collapses to Method (1).
- **Unfavorable δ signs:** If $\delta_0^{(m)} > 0$ or $\delta_1^{(m)} < 0$, the reference-induced terms may cancel or even harm discrimination.

B.7. Limitations: Latent Variables and Empirical Verification

The theoretical framework developed above provides interpretable insights into why reference-based judging methods may improve discrimination. However, several key quantities in the model are *latent variables* that cannot be directly observed or estimated from data.

Unobservability of the usability indicator U . The usability indicator U_j is defined in (16) as the conjunction of two conditions:

- $M_j = 1$: the reference is semantically matched to the target instance;
- $C_j = 1$: the reference answer is correct.

Neither condition is directly observable in practice:

- *Semantic match (M)* is a qualitative judgment that lacks a precise operational definition. While embedding similarity (e.g., cosine similarity) can serve as a proxy, the threshold for “semantic match” is inherently ambiguous.
- *Reference correctness (C)* requires ground-truth labels for all reference answers, which are typically unavailable; indeed, the entire motivation for self-judging is to avoid requiring such labels.

As a consequence, the model parameters $u_y^{(m)} = \Pr(U = 1 \mid Y = y)$ and $\delta_y^{(m)}$ cannot be directly estimated by conditioning on observed values of U .

Indirect verification through aggregate predictions. Although individual latent quantities are unobservable, the model makes testable predictions at the aggregate level:

1. **Relative AUROC ordering:** Under the expected conditions, we predict $\text{AUROC}^{(3)} > \text{AUROC}^{(2)} > \text{AUROC}^{(1)}$.
2. **Scaling with k :** AUROC should improve (or at least not degrade) as the number of reference trials k increases, with diminishing returns governed by the effective sample size k_{eff} .
3. **Proxy-based correlation analysis:** Using embedding similarity as a proxy for M , one can test whether higher-similarity references correlate with improved discrimination.

Role of the theoretical framework. Given these limitations, the primary value of the theoretical analysis lies in:

- *Mechanistic insight:* identifying the key factors (usability, class-specific effects, noise averaging) that govern performance differences;
- *Qualitative predictions:* explaining *why* certain methods should outperform others under specific conditions;
- *Design guidance:* suggesting that improving semantic retrieval quality (to increase $u_1^{(m)}$) and ensuring diverse references (to reduce noise correlation) are promising directions.

Table 6. LLM Retrieval Strategy Comparison (best)

Dataset	Model	Self	Unified	Q-Self	Q-Unified
HaluEval	LLaMA-3.1	71.23	71.48	70.47	71.17
HaluEval	Ministral	67.37	67.62	67.26	67.55
HaluEval	Qwen3	65.53	65.53	65.05	65.05
TruthfulQA	LLaMA-3.1	60.28	59.20	58.15	58.45
TruthfulQA	Ministral	61.88	63.76	63.54	63.93
TruthfulQA	Qwen3	72.28	72.28	72.61	72.61

The framework should be understood as providing interpretive scaffolding rather than a quantitatively precise predictive model. Empirical validation focuses on aggregate predictions (AUROC comparisons, scaling behavior) rather than direct estimation of latent parameters.

C. Retrieval Strategy Analysis

We investigate how different kNN retrieval strategies affect SAGE’s performance. Specifically, we compare four strategies for LLM generation tasks: (1) **Self**: retrieving neighbors using embeddings from the same model based on full question-answer similarity; (2) **Unified**: same as Self but using a unified embedding extractor (Qwen3 for text tasks, MetaCLIP2 for VLM tasks) instead of model-specific embeddings; (3) **Q-Self**: retrieving neighbors based on question similarity only using the same model’s embeddings; (4) **Q-Unified**: question-only retrieval using the unified extractor.

C.1. LLM Retrieval Strategies

Table 6 presents the comparison of different retrieval strategies on LLM generation tasks.

On HaluEval, full input retrieval (Self/Unified) generally outperforms question-only retrieval (Q-Self/Q-Unified). Unified kNN achieves the best or tied-best performance for all three models (71.48%, 67.62%, 65.53%). This suggests that for hallucination detection, answer content provides important semantic signals for finding relevant calibration neighbors.

On TruthfulQA, the pattern differs. Question-only retrieval achieves the best performance for Ministral (Q-Unified: 63.93%) and Qwen3 (Q-Self/Q-Unified: 72.61%), while full input retrieval works best for LLaMA (Self: 60.28%). For factuality assessment, questions with similar topics may share common knowledge requirements, making question similarity a useful proxy for finding relevant references.

C.2. Self vs. Unified Embeddings Across Tasks

Table 7 compares Self kNN and Unified kNN across all task types to assess whether a unified embedding model can match task-specific self-embeddings.

For image classification tasks, Unified kNN consistently outperforms Self kNN across all 6 configurations, with particularly large gains on CIFAR-10 InternVL3.5 (96.95% vs. 92.99%) and ImageNet-1k Qwen3-VL (75.29% vs. 65.71%). For text classification tasks, Unified kNN matches or slightly outperforms Self kNN in all configurations. For VLM tasks, the results are mixed: Unified kNN performs better on COCO for 2 out of 3 models, while Self kNN performs better on Flickr30k for 2 out of 3 models. Overall, Unified kNN achieves equal or better performance in 15 out of 18 configurations across IC, TC, and VLM tasks, demonstrating that SAGE can be effectively deployed with a single embedding model across different generators, eliminating the need for model-specific embedding extraction.

C.3. Visualization of kNN Retrieval

To qualitatively illustrate the differences between embedding models, we visualize the kNN retrieval results for representative query images across four datasets. Each figure shows a query image and its top-9 nearest neighbors retrieved by four different embedding models: Qwen3-VL, InternVL3.5, SAIL-VL2, and MetaCLIP2 (Chuang et al., 2025). Similarity scores are shown below each neighbor image.

Figures 2 and 3 show results for image classification tasks. Despite being trained for different purposes, the unified MetaCLIP2 embedding consistently retrieves visually similar images across both datasets. Figures 4 and 5 show results for

Table 7. Self kNN vs. Unified kNN Comparison

Dataset	Model	Self kNN	Unified kNN
CIFAR-10	InternVL3.5	92.99	96.95
CIFAR-10	Qwen3-VL	94.35	94.54
CIFAR-10	SAIL-VL2	66.93	67.35
ImageNet-1k	InternVL3.5	75.40	79.52
ImageNet-1k	Qwen3-VL	65.71	75.29
ImageNet-1k	SAIL-VL2	68.52	69.06
AG News	LLaMA-3.1	62.44	62.52
AG News	Ministral	64.79	65.31
AG News	Qwen3	71.94	71.94
MMLU	LLaMA-3.1	57.48	57.84
MMLU	Ministral	52.84	52.91
MMLU	Qwen3	68.99	68.99
COCO	InternVL3.5	60.01	59.76
COCO	Qwen3-VL	55.66	55.85
COCO	SAIL-VL2	58.99	61.45
Flickr30k	InternVL3.5	64.16	64.69
Flickr30k	Qwen3-VL	62.60	60.41
Flickr30k	SAIL-VL2	64.50	63.92

image captioning tasks, where visual similarity is particularly important for finding relevant reference captions.

D. Additional Experiments from the Rebuttal

This appendix consolidates the supplementary experiments contributed during the ICML 2026 rebuttal phase, providing the full tables and methodology behind the summarized results in §5.3.

D.1. Stronger Baselines: Full Comparison

The full set of baselines evaluated on TruthfulQA (Qwen3-8B, 817 samples) is reported in Table 8. We include three categories: (i) generation-stage confidence signals (Logprob, Self-consistency over embeddings), (ii) reference-free judge-based methods (Direct, Self-consistency over judge calls), and (iii) reference-based methods (DS2, Random, SAGE).

Table 8. Complete stronger-baselines comparison on TruthfulQA (Qwen3-8B).

Method	AUROC (%)
Self-consistency (embedding)	47.41
Logprob (mean token log-prob)	55.58
Direct	65.34
DS2 (Pang et al., 2025b)	67.10
Self-consistency (judge)	68.58
Random ($k=9$ random refs)	69.02
SAGE ($k=9$ semantic refs, Ours)	72.61

Why generation-stage confidence baselines fail. On TruthfulQA, incorrect answers are often *popular misconceptions* that the generator reproduces consistently across resamples. Self-consistency (embedding) measures the mean cosine similarity between an original answer and five resampled answers; for popular misconceptions, this similarity is high, indistinguishable from the similarity for correct answers, yielding AUROC 47.41% (below chance). Logprob similarly assigns high mean token log-probability to popular wrong answers, achieving only 55.58%. These two baselines illustrate that generation-stage confidence signals alone cannot separate correct from confidently-wrong answers on TruthfulQA, motivating the use of external reference context.

Isolating the contribution of semantic retrieval. The four reference-based or judge-based variants (Direct, Self-consistency-judge, Random, SAGE) form a controlled isolation. **Direct** (65.34%) uses one judge call with no reference.

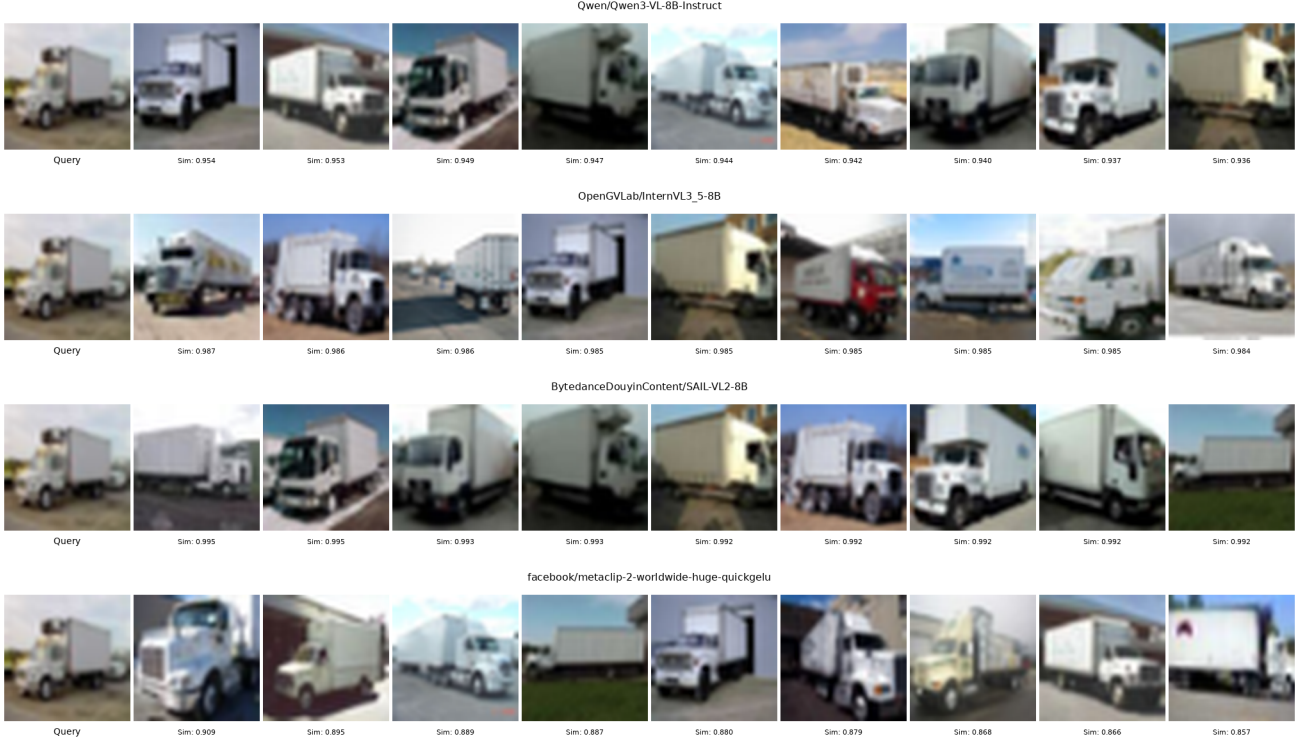


Figure 2. Neighbor retrieval comparison on CIFAR-10. Each row shows the query image and its top-9 nearest neighbors retrieved by different embedding models. Similarity scores are shown below each neighbor image.

Self-consistency (judge) (68.58%) uses k judge calls with no reference, isolating the gain from multi-trial averaging in the absence of context. **Random** (69.02%) uses k judge calls with random references, isolating the marginal effect of *any* external reference context. **SAGE** (72.61%) uses k judge calls with semantically retrieved references, isolating the contribution of relevance. The two informative gaps are Random – Self-consistency (judge) = +0.44 (small gain from context per se) and SAGE – Random = +3.59 (gain from semantic relevance).

D.2. Cross-Model Evaluation: Full Configuration Grid

The main-body summary in §5.3 reports SAGE with Qwen3 embeddings only; here we additionally evaluate LLaMA-3.1 embeddings as the retrieval encoder to assess whether SAGE’s effectiveness is sensitive to the choice of embedding model (Table 9).

Table 9. Full cross-model evaluation on TruthfulQA. SAGE is evaluated with both LLaMA-3.1 embeddings (L-emb) and Qwen3 embeddings (Q-emb). Encoder choice changes SAGE AUROC by less than 1 pp in every cross-model configuration.

Generator	Judge	Direct	Random	SAGE (L-emb)	SAGE (Q-emb)
Qwen3	Qwen3 (same)	65.34	69.02	—	72.61
LLaMA-3.1	Qwen3 (cross)	65.02	68.11	70.43	71.32
LLaMA-3.1	Ministral (cross)	57.20	58.46	60.06	60.34

Two robustness observations follow. *First, cross-encoder robustness.* The L-emb and Q-emb columns differ by at most 0.89 pp (LLaMA→Qwen3) and 0.28 pp (LLaMA→Ministral), indicating that SAGE’s gain is insensitive to the specific embedding model used for retrieval. *Second, cross-judge robustness.* The ranking SAGE > Random > Direct holds in all three configurations, and the same-model vs. cross-model SAGE gap is only 1.29 pp (71.32% vs. 72.61%), supporting the theoretical claim (Eq. 10) that SAGE’s mechanism does not require the generator and judge to share a model. The Ministral configuration produces lower absolute AUROC across all methods, reflecting Ministral’s weaker judging ability, but SAGE’s *relative* gain over Direct (+3.14 pp) is preserved.

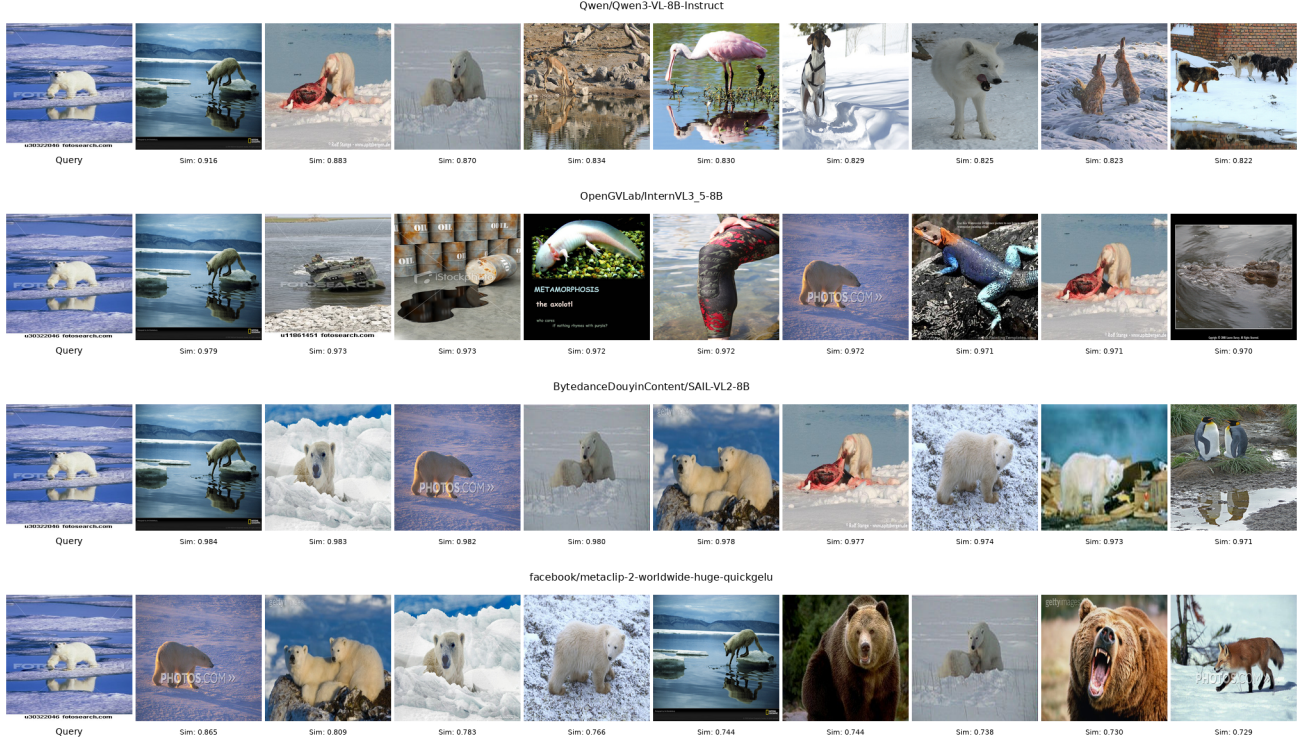


Figure 3. Neighbor retrieval comparison on ImageNet-1k. Each row shows the query image and its top-9 nearest neighbors retrieved by different embedding models. Similarity scores are shown below each neighbor image.

D.3. Human-Evaluation Alignment Study

To assess whether the model-based ground-truth labels used in our generation experiments reflect human judgments, we re-annotated 817 TruthfulQA samples with four independent human raters (denoted A, B, C, D). Inter-rater agreement is moderate (Fleiss’ $\kappa = 0.552$), reflecting the inherent subjectivity of correctness judgments on TruthfulQA, where many questions admit multiple defensible answers. The per-annotator and 80B-judge correctness rates are summarized in Table 10.

Table 10. Annotator-level correctness rates on the 817-sample TruthfulQA subset.

	Annotator A	Annotator B	Annotator C	Annotator D	80B Judge
% rated correct	43.8	73.3	43.1	53.2	44.9

We then evaluated SAGE, Random, and Direct under seven distinct ground-truth definitions: each annotator individually, two majority-vote thresholds ($\geq 3/4$ and $\geq 2/4$ correct), and the 80B judge baseline used in the main experiments. Table 11 reports the resulting AUROC values.

The robustness of the ranking under every individual annotator (despite substantial inter-rater disagreement) and under both majority thresholds indicates that SAGE’s advantage is not an artifact of the 80B judge’s particular notion of correctness. The narrow band of SAGE–Direct gaps (+4.49 to +7.27 pp) further suggests that the absolute magnitude of the gain is also robust to the choice of correctness criterion.

D.4. Sensitivity Analysis on Reference Count k

We sweep the reference count k over $\{1, 3, 5, 7, 9, 15, 21\}$ on TruthfulQA with Qwen3-8B (Table 12).

Three observations follow. First, even $k=1$ produces a +3.50 pp improvement over Direct, indicating that a single semantically relevant neighbor already serves as a meaningful calibration anchor. Second, AUROC is stable across $k=3$ through $k=9$ (72.23–72.61%), suggesting the gain plateaus once a small number of useful neighbors has been retrieved,

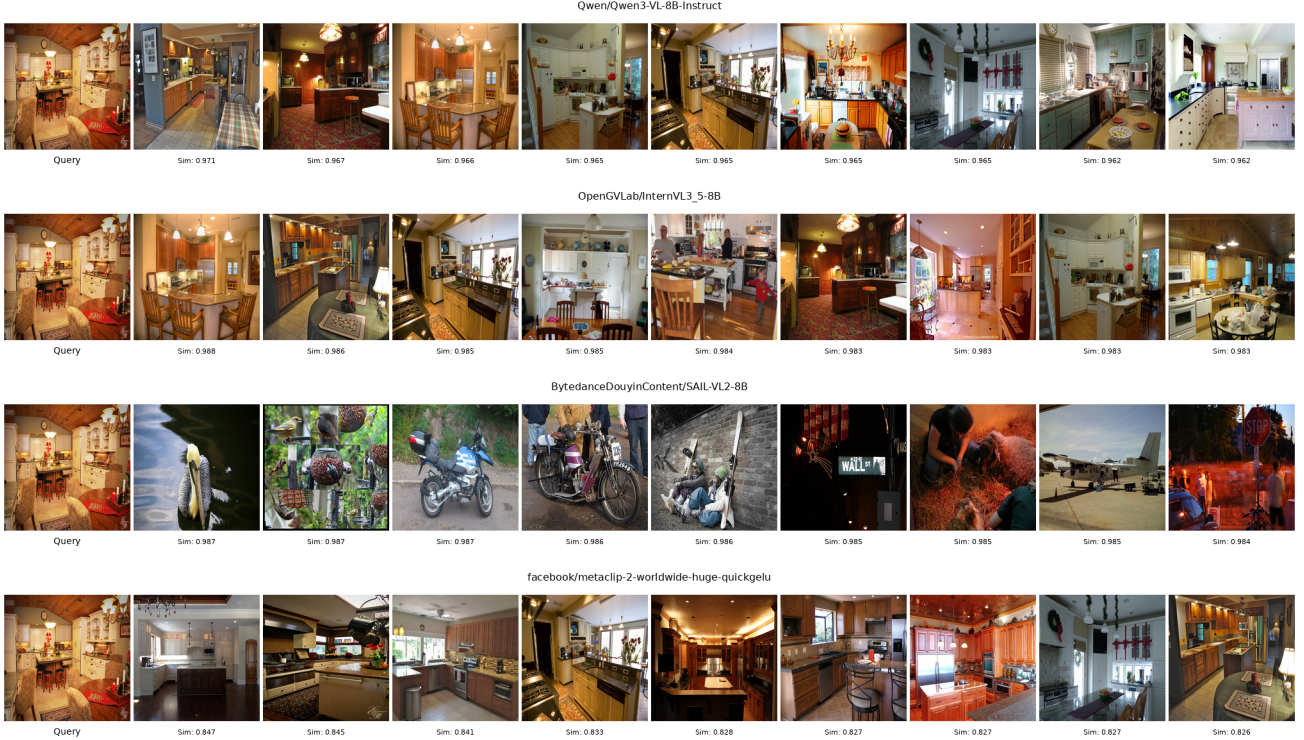


Figure 4. Neighbor retrieval comparison on COCO-Caption. Each row shows the query image and its top-9 nearest neighbors retrieved by different embedding models. Similarity scores are shown below each neighbor image.

Table 11. Error-detection AUROC (%) under seven ground-truth definitions on TruthfulQA (Qwen3-8B). The ranking $\text{SAGE} > \text{Random} > \text{Direct}$ holds in every row, with $\text{SAGE} - \text{Direct}$ gaps ranging from +4.49 to +7.27 pp.

Ground Truth	SAGE	Random	Direct	$\text{SAGE} - \text{Random}$	$\text{SAGE} - \text{Direct}$
Annotator A	68.41	66.40	63.23	+2.01	+5.18
Annotator B	74.08	71.86	68.58	+2.22	+5.50
Annotator C	68.56	66.29	63.09	+2.27	+5.48
Annotator D	71.41	69.63	66.92	+1.78	+4.49
Majority ($\geq 3/4$)	69.55	67.42	64.50	+2.13	+5.06
Majority ($\geq 2/4$)	70.62	68.69	64.49	+1.93	+6.13
80B Judge	72.61	69.02	65.34	+3.59	+7.27

consistent with the effective sample size $k_{\text{eff}} \approx 2$ at $k=9$ implied by the noise correlation $\hat{\rho} \approx 0.45$ (Appendix D.6). Third, SAGE outperforms Random at every k , isolating the contribution of semantic relevance from the contribution of simple multi-trial averaging. Practitioners constrained by judging cost may choose $k=3$ to capture most of the gain at roughly $3 \times$ Direct cost.

D.5. Runtime Reliability Proxy: Per-Sample Score Variance

In response to a follow-up question about lightweight, label-free signals for flagging unreliable SAGE predictions at inference time, we propose the variance of the k neighbor-trial scores as such a proxy: high variance reflects neighbor disagreement and unreliable retrieval; low variance reflects stable, consistent calibration. The proxy requires no additional model calls beyond the standard SAGE pipeline.

Within-configuration behavior. On TruthfulQA (Qwen3-8B, $k=9$), splitting the 817 samples into three groups by per-sample score variance yields Table 13. SAGE AUROC monotonically decreases with variance (Pearson $r = -0.804$ between per-sample variance and within-group SAGE AUROC), yet SAGE outperforms Direct in every variance group; within a given task and model, a hard fallback to Direct is rarely needed.

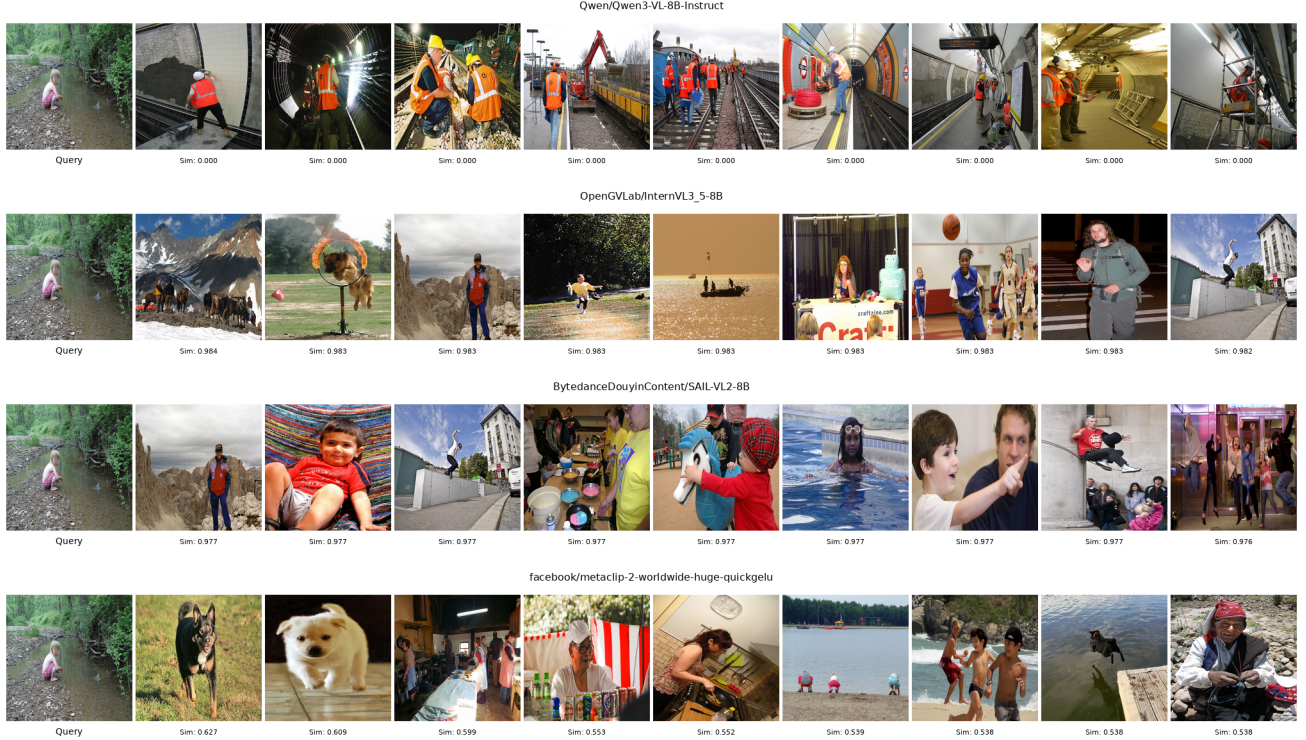


Figure 5. Neighbor retrieval comparison on Flickr30k. Each row shows the query image and its top-9 nearest neighbors retrieved by different embedding models. Similarity scores are shown below each neighbor image.

Table 12. k -sensitivity on TruthfulQA (Qwen3-8B). SAGE outperforms Direct (65.34%) at every k and outperforms Random at every k .

k	SAGE AUROC	Random AUROC	SAGE–Random
1	68.84	65.17	+3.68
3	72.23	67.98	+4.25
5	72.24	68.90	+3.33
7	72.32	69.08	+3.24
9	72.61	69.02	+3.59
15	71.75	69.61	+2.14
21	71.64	69.80	+1.85

Cross-configuration behavior. Aggregating the mean per-sample score variance across seven configurations spanning TruthfulQA and CIFAR-10 reveals a clean separation (Table 14): every configuration where SAGE outperforms Direct has mean variance ≤ 0.51 , while every configuration where SAGE underperforms has mean variance ≥ 0.57 .

Practical workflow. The variance proxy suggests different actions depending on its magnitude:

- **Variance** < 0.50 : use SAGE with confidence.
- **Variance** 0.50 – 0.70 : likely suboptimal retrieval; switch retrieval strategy (e.g., from question-only to full-input embeddings, or to a unified external encoder). This recovers SAGE’s advantage on TruthfulQA/LLaMA (Tables 6–7).
- **Variance** > 1.0 : the judge model itself produces unreliable scores on the task (as with CIFAR-10/SAIL-VL2); consider using a stronger judge model.

The proxy does not flag the ceiling-effect regime (e.g., COCO/Qwen3-VL with $\sim 85\%$ base accuracy), where both δ_1 and δ_0 approach zero and no method meaningfully improves detection. In this regime SAGE matches Direct without harming it.

Table 13. Within-configuration variance stratification on TruthfulQA (Qwen3-8B, $k=9$). SAGE outperforms Direct in every group despite the decline in absolute AUROC with variance.

Variance Group	Range	N	SAGE AUROC	Direct AUROC
Low	[0.00, 0.40]	386	74.06	71.57
Medium	[0.40, 0.69]	138	62.21	55.83
High	[0.69, 1.78]	293	63.11	57.59

Table 14. Cross-configuration variance signal. Configurations with elevated mean per-sample variance (≥ 0.57) coincide with SAGE underperforming Direct.

Configuration	SAGE AUROC	Mean Score Variance	SAGE vs. Direct
CIFAR-10 / Qwen3-VL (Self-emb)	94.41	0.27	SAGE best
CIFAR-10 / Qwen3-VL (MetaCLIP)	94.63	0.34	SAGE best
TruthfulQA / Qwen3	72.61	0.38	SAGE best
TruthfulQA / Ministral	61.67	0.51	SAGE best
TruthfulQA / LLaMA-3.1	60.19	0.57	Direct best
CIFAR-10 / SAIL-VL2 (Self-emb)	66.02	1.52	Direct best
CIFAR-10 / SAIL-VL2 (MetaCLIP)	66.36	1.63	Direct best

D.6. Empirical Estimates of Latent Theoretical Quantities

From experimental scores on TruthfulQA (Qwen3-8B, $k=9$), we estimate the latent quantities in the AUROC closed form (Eq. 10): reference correctness $\Pr(C=1)$, score shifts δ_1 and δ_0 , and noise correlation ρ . Because kNN retrieval ensures the semantic match indicator M is approximately 1 on average, we use C as a proxy for the usability indicator $U=M \wedge C$ throughout.

Reference correctness rate and controlled ablation. The pool-level correctness rate is $\Pr(C=1) = 0.449$, equal to the generator’s base accuracy on TruthfulQA. To test the predicted monotonic dependence of AUROC on $\Pr(C=1)$, we retrieve $k=9$ neighbors as usual and then randomly drop a fraction of *correct* neighbors before averaging. Table 15 shows AUROC decreases monotonically as $\Pr(C=1)$ falls, matching the prediction from Eq. (10) that $\Pr(C=1)$ scales u_y and thus the calibration bonus $\delta_1 u_1 - \delta_0 u_0$.

Table 15. Reference-quality ablation. Manipulating effective $\Pr(C=1)$ by selectively dropping correct neighbors causes AUROC to decrease monotonically. The “Only correct” row gives the oracle upper bound at $\Pr(C=1)=1$.

Condition	Effective $\Pr(C=1)$	AUROC (%)
Only correct refs (oracle)	1.000	81.26
SAGE (all refs)	0.449	72.61
Drop 20% correct refs	0.395	70.63
Drop 40% correct refs	0.329	69.00
Drop 60% correct refs	0.249	66.74
Direct	—	65.34

Score shifts δ_1, δ_0 . We stratify all neighbor-trial scores by target correctness Y and reference correctness C , then estimate $\hat{\delta}_1 = \mathbb{E}[S \mid Y=1, C=1] - \mathbb{E}[S \mid Y=1, C=0]$ and $\hat{\delta}_0$ analogously. The estimates are $\hat{\delta}_1 = +0.47$ and $\hat{\delta}_0 = -0.16$, matching the theoretical signs predicted in §4.2: usable references raise scores for correct targets and lower them for incorrect targets.

Noise correlation ρ . We estimate ρ as the average pairwise Pearson correlation of within-sample score residuals $\epsilon_j = S_j - \mathbb{E}[S \mid Y, C_j]$ across the k neighbor trials. The estimate is $\hat{\rho} \approx 0.45$, yielding an effective sample size $k_{\text{eff}} = k/(1 + (k-1)\rho) \approx 2$ at $k=9$. This is consistent with the saturation observed in the k -ablation (Appendix D.4): once k exceeds a few effectively independent trials, additional neighbors provide diminishing returns.